

A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

ABSTRACT. A countable discrete group is MF if it embeds in the unitary group of a matrix quotient $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$; equivalently, if it admits finite-dimensional approximate unitary representations, asymptotically multiplicative in operator norm, that separate every nontrivial element from the identity. We construct an explicit finitely presented group E that is not MF, refuting the conjecture that every countable group is MF.

More generally, let Γ be a finitely presented property-(T) group with a proper injective endomorphism α . From (Γ, α) we build a finitely presented group with a nontrivial central involution w such that $\Theta(w) = 1$ for every homomorphism Θ to $\mathcal{U}(\mathcal{Q})$. In every operator-norm asymptotic representation, one-sided conjugation preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm; if $\Theta(w) \neq 1$, compression to the (-1) -spectral projection of $\Theta(w)$ contradicts this. Taking $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and $\alpha(v, A) = (2v, A)$ gives E , on eight generators and forty-one relators.

The group E is sofic, hence hyperlinear, so neither soficity nor hyperlinearity implies the MF property; and its reduced C^* -algebra $C_r^*(E)$ is separable and stably finite but not MF.

2020 *Mathematics Subject Classification*. Primary 46L05; Secondary 20F05, 22D25, 20E26.

Key words and phrases. MF groups, approximate unitary representations, property (T), group C^* -algebras, Clifford groups.

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1. INTRODUCTION

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. We prove that the answer is no: there is an explicit finitely presented group, on eight generators and forty-one relators, that is not MF. Residually finite groups satisfy the conjecture through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP]. Throughout, $[g, h] = ghg^{-1}h^{-1}$ denotes the commutator of two group elements. The additive commutator of two operators is written $[x, y]_- = xy - yx$, so that the two brackets are never confused where a group element and its matrix image occur together.

The result.

Theorem A (an explicit finitely presented non-MF group). *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 6.1, and put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

The defining relations make w central; together with $c^2 = 1$, centrality implies $w^2 = 1$. Then the following statements hold:

- (1) *$w \neq 1$ in E , so w is a central involution;*
- (2) *for every sequence (d_n) of positive integers and every homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$ into the unitary group of the corresponding matrix quotient, one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C_{\max}^(E)$ nor $C_r^*(E)$ is an MF C^* -algebra. A Tietze reduction presents the same group on six generators with thirty-two relators (Section S1).*

The proof is in Section 6.

Outline of the proof. An inclusion between two objects of equal finite size is an equality.

$$(1) \quad \begin{array}{ccc} X \subseteq Y, |X| = |Y| & V \subseteq W, \dim V = \dim W & \begin{array}{c} p \leq q, p \sim q \\ \text{in a finite } C^*\text{-algebra} \end{array} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

Here $p \sim q$ is Murray–von Neumann equivalence, and a C^* -algebra is *finite* when every isometry in it is unitary. The three columns give the arguments for finite quotients, for finite-dimensional representations, and for $\mathcal{Q}$, in that order. Operator norm is the approximation norm throughout, and the normalized Hilbert–Schmidt norm enters only as the Hilbert-space norm on which property (T) acts.

Let π be a homomorphism to a finite group and let $t\Gamma t^{-1} \subseteq \Gamma$. Then $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$: the left side is a conjugate of the right side contained in it, and the two are finite of the same cardinality. So an element c centralizing Γ has $\pi(tct^{-1})$ centralizing $\pi(\Gamma)$, and every commutator $[tct^{-1}, a]$ with $a \in \Gamma$ lies in $\ker \pi$.

Every finite-dimensional representation π of the ambient group over every field gives the same conclusion, with the commutant in place of the subgroup. Let C be the commutant of $\pi(\Gamma)$ in $\text{End}_k(V)$. The inclusion $t\Gamma t^{-1} \subseteq \Gamma$ implies $C \subseteq \pi(t)C\pi(t)^{-1}$, the two sides have equal finite dimension, and therefore

$$C = \pi(t)C\pi(t)^{-1}$$

(Theorem 3.3 and Remark S7.2). Since $\pi(c) \in C$, also $\pi(tct^{-1}) \in C$, so it commutes with every $\pi(a)$, $a \in \Gamma$.

Regard $M_d(\mathbb{C})$ as the Hilbert space $L^2(M_d(\mathbb{C}), \text{tr}_d)$, on which conjugation by a unitary acts unitarily. For unitaries u_1, u_2 , uniformly in d ,

$$\|\text{Ad } u_1 - \text{Ad } u_2\|_{B(L^2(M_d(\mathbb{C}), \text{tr}_d))} \leq 2 \|u_1 - u_2\|,$$

so an operator-norm asymptotic representation induces one by conjugation on the Hilbert–Schmidt spaces.

Let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and let $t, c \in H$ satisfy

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

In a norm ultraproduct B_ω of the blocks, the classes $\pi(g) = [\text{Ad } U_n(g)]_\omega$ define a unitary representation of H . By property (T), the spectrum of the Hermitian average h of the generators is bounded away from 1 off the invariant vectors, so the spectral projection of h at 1 lies in B_ω and projects onto the $\iota(\Gamma)$ -invariant vectors; call it P . Writing $V = \pi(t)$, the inclusion above implies $P \leq VPV^*$; the two projections are Murray–von Neumann equivalent, and B_ω contains no proper isometry, so $P = VPV^*$. Hence conjugation by t preserves the asymptotic commutant of $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm (Theorem 3.4).

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for $a \in \Gamma$. Since c commutes with $\iota(\Gamma)$, its lifts lie in that asymptotic commutant, so by Theorem 3.4 $\|[U_n(d), U_n(\iota(\gamma))]\|_2 \rightarrow 0$ for every $\gamma \in \Gamma$. Consequently, in every operator-norm asymptotic representation (U_n) of H ,

$$\|U_n(u) - 1\|_2 \rightarrow 0.$$

Suppose now that $w = u^2$ is central in H with $w^2 = 1$, and let Θ be a homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, so

$$q = \frac{1}{2}(1 - \Theta(w))$$

is a nonzero projection commuting with $\Theta(H)$. Choosing unitary lifts of Θ and rounding the corresponding coordinate averages to genuine projections represents q by projections $q_n \in M_{d_n}(\mathbb{C})$, nonzero for infinitely many n and

asymptotically commuting with the lifts. Compressing to q_n , and giving each corner its own normalized trace, yields a second operator-norm asymptotic representation of H , one in which w goes to -1 . In that representation the display above gives $\|U_n(u)^2 - 1\|_2 \rightarrow 0$, while $\|-1 - 1\|_2 = 2$. Hence $\Theta(w) = 1$ for every such Θ , and if $w \neq 1$ then H is not MF (Theorem 4.1).

Let Γ be a finitely presented Kazhdan group with a proper injective endomorphism α , let $a \in \Gamma \setminus \alpha(\Gamma)$, and adjoin t and c subject to $t\gamma t^{-1} = \alpha(\gamma)$, $c^2 = 1$ and $[c, \gamma] = 1$. In the direct limit of α the cosets $t\Gamma$ and $a\Gamma$ are distinct, since $t^{-1}at \in \Gamma$ would give $a \in \alpha(\Gamma)$. Let the ambient group permute the generators of a Clifford group indexed by that coset space, and send c to the generator at the base coset; then d and ada^{-1} map to the generators at those two cosets. Distinct Clifford generators anticommute, so u maps to a product of two anticommuting involutions and $w = u^2$ to -1 (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I)$$

gives the group E of Theorem A.

Only the subgroup $\iota(\Gamma)$ is assumed to have property (T). The group E surjects onto $\mathbb{Z}$, by sending t to 1 and every other generator to 0, so E itself does not have property (T).

The reduced group C^* -algebra $C_r^*(E)$ is separable and stably finite but not MF (Theorem C). The Clifford quotient W used to prove $w \neq 1$ is itself finitely generated, sofic and not MF, so neither soficity nor hyperlinearity implies the MF property (Theorem D); the group E is sofic as well (Theorem E), so a finitely presented group already separates the two properties. The quotient $E/\langle w \rangle$ is also not MF, so $\{1, w\}$ is a proper subgroup of the MF residual of E (Theorem B). The non-MF argument implies neither soficity nor nonsficity, for either group (Section S14).

Relation to prior work. The corresponding C^* -algebra question is older. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^* -algebra was known not to be MF [BK, CDE]. Their natural candidate was $C_r^*(F_2)$, which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently supplied abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group C^* -algebra fails to be MF [Sh].

In his 2018 ICM address, Thom asked for which Schatten p -norms there exist non-approximable groups [Th, Question 3.11]. The case $p = 2$ was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky–Oppenheim [LuO], and the case $p = 1$ by Bachner–Dogon–Lubotzky [BDL]. Three approximation problems were left by that work: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), and MF (unitaries in operator norm). The first was answered negatively

in 2026: a nonsofic group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. The present paper answers the third negatively; at the time of writing the hyperlinear problem is open. Our construction begins, as those arguments do, with a relation $t\Gamma t^{-1} \subseteq \Gamma$ for a property-(T) subgroup Γ . The approximation here is by unitaries in operator norm, not by permutations in the Hamming metric, and the rigidity it supplies is that conjugation by t preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm, in every operator-norm asymptotic representation. Throughout, “MF” is the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE], in the form used by Shulman and by Bachner–Dogon–Lubotzky [Sh, BDL]; Section 2 fixes it and its equivalent formulations.

The closest analogue is the recent work on nonsofic groups. One-sided conjugation of a property-(T) subgroup, together with an element centralizing that subgroup, appears in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the group acts by conjugation on the Hilbert–Schmidt space of each matrix block, and finiteness of the norm ultraproduct as a C^* -algebra forces the projection onto the $\iota(\Gamma)$ -fixed subspace to equal its unitary conjugate (Theorem 3.4, and Section S4 for a finite-stage form).

Slofstra’s hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3], and his quotients by the elements sent to the identity by all exact, respectively by all approximate, finite-dimensional representations [Slo17, Definitions 2.5–2.7] are the exact and the approximate analogues of the MF residual used below. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz’s marked BS(2,3) example is the context for the cyclic comparison of Section S8 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in Section S6 [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5], and handle a finite normal subgroup by a character-isotypic corner [BDL, proof of Proposition 1.6]; Theorem S6.3 combines that corner with one-sided conjugation of a property-(T) subgroup.

Bekka and Bekka–Valette study property (T) through the conjugation representation $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt space [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

The group property is weaker than the algebra property. For the algebra one asks for an embedding $C_r^*(E) \hookrightarrow \mathcal{Q}$ of C^* -algebras; for the group, only for an injective homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$, which need not extend to the

algebra. Failure of MF for a group C^* -algebra therefore leaves open whether the group itself embeds. For E , Theorem A rules out both.

2. MATRIX QUOTIENTS

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. A C^* -homomorphism is unital only when this is stated; in particular an MF embedding may be nonunital. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$. For a nonzero projection $q \in M_r(\mathbb{C})$ we equip the corner $qM_r(\mathbb{C})q$ with its own normalized trace, and where the trace has to be named we write $\|x\|_{2,\text{tr}_r}$ for the same quantity. The unnormalized trace is $\text{Tr} = r \cdot \text{tr}_r$ on $M_r(\mathbb{C})$. Convergence in normalized Hilbert–Schmidt norm is always asserted of sequences of matrices, never of group elements. We use without comment: $\|x\|_2 \leq \|x\|$; $|\text{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v .

Given a sequence $(d_n)_{n \in \mathbb{N}}$ of positive integers, write

$$(2) \quad \mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}),$$

where the product is the bounded C^* -product and $\bigoplus_n$ is the c_0 -sum, the ideal of sequences with $\|x_n\| \rightarrow 0$; equivalently, $\mathcal{Q}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$. Write $q: \prod_n M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ for the quotient map. Following Blackadar–Kirchberg [BK], a separable C^* -algebra is MF if it embeds into an algebra of the form (2).

The following lifting lemmas are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 2.1 (lifting). *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

Proof. Lift u to a bounded sequence (x_n) ; unitarity of $q((x_n))$ gives $\|x_n^*x_n - 1\| \rightarrow 0$. Once $\|x_n^*x_n - 1\| \leq \frac{1}{2}$, the matrix x_n is invertible and the polar correction $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \leq 2\|x_n\|\|x_n^*x_n - 1\| \rightarrow 0$ by continuous functional calculus; set $u_n = 1$ at the finitely many remaining indices. $\square$

The unitaries of $\mathcal{Q}$ admit a direct description. Write

$$(3) \quad \mathcal{N} = \{(u_n) \in \prod_n \mathcal{U}(d_n) : \|u_n - 1\| \rightarrow 0\},$$

a normal subgroup of $\prod_n \mathcal{U}(d_n)$.

Lemma 2.2 (unitaries in the quotient). *For every sequence (d_n) , with $\mathcal{N}$ as in (3), the map*

$$\kappa_{(d_n)}: \prod_n \mathcal{U}(d_n) / \mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. A homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ is therefore equivalent to the following data: a sequence of unitaries for each element

of H , multiplicative up to operator-norm errors tending to 0, taken modulo sequences converging to 1.

Proof. The kernel of $(u_n) \mapsto q((u_n))$ is exactly $\mathcal{N}$, so the induced map is injective. Surjectivity is Lemma 2.1. $\square$

An *operator-norm asymptotic representation* of H is a sequence of maps $U_n: H \rightarrow \mathcal{U}(d_n)$ such that $\|U_n(g)U_n(h) - U_n(gh)\| \rightarrow 0$ for all $g, h \in H$. Taking $g = h = 1$ gives $\|U_n(1) - 1\| \rightarrow 0$ automatically, since $U_n(1)$ is unitary. By Lemma 2.2, choosing unitary lifts $U_n(g)$ of $\Theta(g)$ turns a homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ for some sequence of dimensions.

Proposition 2.3 (Equivalent formulations of the MF property). *For every countable group, the C^* -algebra definition above is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

Section S3 defines the approximate unitary representations and proves the proposition. Two stronger conventions occur in the literature and are not used here: in [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element, and the purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

3. ONE-SIDED CONJUGATION IN MATRIX MODELS

Theorem 3.3 uses the middle implication of (1) for the commutant C of the base inside a finite-dimensional endomorphism algebra, where $C \subseteq \pi(t)C\pi(t)^{-1}$ and the two sides have equal finite dimension. The proof of Theorem 3.4 uses the right-hand implication for the projection P onto the Kazhdan fixed space inside a norm ultraproduct: there $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$ gives $P \leq VPV^*$, the two projections are Murray–von Neumann equivalent, and the ultraproduct is finite, so $P = VPV^*$.

Lemma 3.1 (comparison in a finite algebra). *Let A be a finite unital C^* -algebra and let $p, q \in A$ be projections with $p \leq q$ and $p \sim q$. Then $p = q$.*

Proof. Let $r \in A$ satisfy $r^*r = q$ and $rr^* = p$. From $r^*r = q$ we get $r = rq$. Since $p \leq q$ we have $qp = p$, so

$$pr = rr^*r = rq = r, \quad qr = q(pr) = (qp)r = pr = r,$$

and $r = qr$ as well. Hence the cross terms in $\sigma^*\sigma$ vanish for $\sigma = r + (1 - q)$, and $\sigma^*\sigma = q + (1 - q) = 1$, so σ is an isometry. Finiteness makes σ unitary, and $\sigma\sigma^* = p + (1 - q) = 1$ gives $p = q$. $\square$

Lemma 3.2 (finiteness of a norm quotient of matrix blocks). *Let (K_n) be finite-dimensional Hilbert spaces and let*

$$B_c = \prod_n B(K_n) / \bigoplus_n B(K_n)$$

be the quotient of the bounded family algebra by the c_0 -sum. Then B_c is a finite C^ -algebra: every $\sigma \in B_c$ with $\sigma^* \sigma = 1$ satisfies $\sigma \sigma^* = 1$.*

Proof. Lift σ to a bounded family (σ_n) . Then $\|\sigma_n^* \sigma_n - 1\| \rightarrow 0$, so for all large n the Gram defect is at most $\frac{1}{2}$ and $\sigma_n^* \sigma_n$ is invertible; each σ_n acts on a finite-dimensional space, so polar correction replaces it by a unitary $w_n = \sigma_n (\sigma_n^* \sigma_n)^{-1/2}$ with $\|\sigma_n - w_n\| \rightarrow 0$. Hence σ is the class of (w_n) , and $w_n w_n^* = 1$ at every such n gives $\sigma \sigma^* = 1$. $\square$

Let ω be a free ultrafilter on $\mathbb{N}$, let K_ω be the Hilbert-space ultraproduct of the K_n , and let $B_\omega = \prod_\omega B(K_n)$ be the norm ultraproduct, acting on K_ω by $[A_n]_\omega [\xi_n]_\omega = [A_n \xi_n]_\omega$. The proof above applies to B_ω with “for all large n ” replaced by “on a set belonging to ω ”, so B_ω is finite too. Its action on K_ω is faithful: if $A = [A_n]_\omega \neq 0$ then $\lim_\omega \|A_n\| = \delta > 0$, and unit vectors ξ_n with $\|A_n \xi_n\| \geq \|A_n\| - 1/n$ give $\|A[\xi_n]_\omega\| = \delta$. Hence for projections $P, Q \in B_\omega$ the inclusion $\text{ran } P \subseteq \text{ran } Q$ makes QP and P act identically on K_ω , so $QP = P$, that is, $P \leq Q$; the converse is immediate.

The exact case.

Theorem 3.3 (the finite-dimensional case). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi: H \rightarrow \text{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such π sends the entire subgroup $\mathfrak{D}(H, \Gamma)$ of (10) to the identity.

Proof. Let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\text{End}_k(V)$, and write $\Phi(x) = \pi(t)x\pi(t)^{-1}$ for conjugation by $\pi(t)$, a linear automorphism of $\text{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and put $y = \pi(t)^{-1}x\pi(t)$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = \pi(t)^{-1}\pi(t\gamma t^{-1})x\pi(t) = \pi(t)^{-1}x\pi(t\gamma t^{-1})\pi(t) = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(4) \quad C = \Phi(C) = \pi(t)C\pi(t)^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ , so $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$ by (4). Since $a \in \Gamma$, the operator $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$ and

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1.$$

Section S7 proves the subgroup-valued assertion from the same calculation. $\square$

In E , every finite-dimensional linear representation over every field, and hence every homomorphism to a finite group, maps the distinguished word to the identity (Corollary S7.1).

An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not invariant subspaces. Operator-norm control keeps the adjoint actions almost multiplicative on the Hilbert–Schmidt spaces, and the finite-stage proof of Section S4 works with the spectral subspace $\text{ran } P_n$ of a Kazhdan average above a fixed threshold and with $Q_n = \text{Ad } U_n(s) P_n \text{Ad } U_n(s)^*$: there $\|(1 - Q_n)P_n\| \rightarrow 0$ and $\text{rank } P_n = \text{rank } Q_n$, hence $\|(1 - P_n)Q_n\| \rightarrow 0$. Property (T) cannot be dropped: the group E_{BS} of Section S8 has cyclic base, its distinguished word is trivial in every finite-dimensional representation over every field, and some homomorphism $E_{\text{BS}} \rightarrow \mathcal{U}(\mathcal{Q})$ maps that word to a nontrivial element.

Theorem 3.4 (one-sided conjugation preserves asymptotic commutants). *Let Γ and H be groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, let $s \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

Let $U_n: H \rightarrow \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and asymptotically centralizes $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm,

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma)) x_n\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma),$$

then the conjugated sequence does as well:

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma).$$

Proof. Suppose the conclusion fails: there are $\gamma_0 \in \Gamma$, $\delta > 0$, and an infinite set $I \subseteq \mathbb{N}$ with $\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma_0))]\|_2 \geq \delta$ for $n \in I$. Fix a free ultrafilter ω on $\mathbb{N}$ with $I \in \omega$.

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ and let each group element act on it by conjugation, $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$. Operator-norm almost multiplicativity of U_n makes $g \mapsto \text{Ad } U_n(g)$ an operator-norm asymptotic representation on these Hilbert spaces.

The ultraproduct. Let K_ω and $B_\omega = \prod_\omega B(K_n)$ be as in Section 3; write $[\cdot]_\omega$ for classes in either. The action of B_ω on K_ω is faithful, B_ω is finite in the sense fixed above — every isometry in it is a unitary, a condition on the algebra

and not on its dimension, which is infinite — and an inclusion of ranges in K_ω is an inequality of projections in B_ω . Asymptotic multiplicativity makes

$$\pi: H \longrightarrow \mathcal{U}(B_\omega), \quad \pi(g) = [\text{Ad } U_n(g)]_\omega,$$

a homomorphism.

The projection onto the fixed subspace. Property (T) provides a finite symmetric generating set $S \subseteq \Gamma$ with $1 \in S$ and a Kazhdan constant $\kappa \in (0, 1]$ for the pair (S, κ) . Let h be the Hermitian part of $\frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$ and let $\text{Fix} \subseteq K_\omega$ be the subspace of vectors fixed by every $\pi(\iota(\gamma))$. Both Fix and its orthocomplement are invariant under each unitary $\pi(\iota(s'))$, hence under h . On Fix , $h = 1$. The restriction of $\pi \circ \iota$ to $\text{Fix}^\perp$ has no nonzero invariant vector, so a unit vector $\xi \in \text{Fix}^\perp$ satisfies $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$ for some generator. For that generator $\text{Re}\langle \pi(\iota(s'))\xi, \xi \rangle \leq 1 - \kappa^2/2$, while every other term is at most 1. Averaging gives $\langle h\xi, \xi \rangle \leq 1 - \kappa^2/(2|S|)$. Consequently

$$\text{sp}(h) \subseteq \left[-1, 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection P of h at the isolated point 1 belongs to B_ω by continuous functional calculus and projects onto Fix .

The two projections agree. Let $V = \pi(s)$ and $Q = VPV^*$. Since $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$, every vector of Fix is fixed by each $\pi(s\iota(\gamma)s^{-1})$; and η is fixed by all $\pi(s\iota(\gamma)s^{-1})$ exactly when $V^*\eta \in \text{Fix}$. Hence $\text{ran } P = \text{Fix} \subseteq V\text{Fix} = \text{ran } Q$, so $P \leq Q$. The element $r = V^*Q$ satisfies $r^*r = Q$ and $rr^* = V^*QV = P$, so $P \sim Q$. Since B_ω is finite, Lemma 3.1 gives $Q = P$.

Conclusion. Let $\xi = [\xi_n]_\omega \in K_\omega$, where $\xi_n \in K_n$ is the vector of x_n ; the uniform operator-norm bound makes ξ well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each $\pi(\iota(\gamma))$ fixes ξ . Thus $\xi \in \text{Fix}$. Since $Q = P$, we obtain $V\xi \in V\text{Fix} = \text{ran } Q = \text{ran } P = \text{Fix}$. The vector of $U_n(s)x_nU_n(s)^*$ is $\text{Ad } U_n(s)\xi_n$, so, again by unitary invariance,

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]\|_2 = \|(\text{Ad } U_n(\iota(\gamma)) - 1) \text{Ad } U_n(s)\xi_n\| \longrightarrow 0$$

along ω , for every $\gamma \in \Gamma$. For $\gamma = \gamma_0$ this contradicts $I \in \omega$. $\square$

Remark 3.5 (the finite-stage proof). Section S4 proves the theorem again at finite stages. Conditional on the spectral-gap estimate stated there, every bound in that proof is an explicit inequality in κ , $|S|$, the uniform bound on (x_n) , and the multiplicative defects of U_n .

4. THE CENTRAL INVOLUTION OBSTRUCTION

Let w be a central involution of H and let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, the projection $\frac{1}{2}(1 - \Theta(w))$ is nonzero, and compressing to it gives a matrix model of H in which w equals -1 and each block has its own normalized trace.

A bound on a commutator in normalized Hilbert–Schmidt norm does not by itself yield an operator-norm obstruction: padding an asymptotic representation with an identity block leaves every operator-norm quantity unchanged and sends every normalized Hilbert–Schmidt distance to zero (Section S14). On the corner $q_n M_{d_n}(\mathbb{C}) q_n$ the operator norm is the restriction of the operator norm, so the compressed and polar-corrected maps remain operator-norm asymptotically multiplicative, while the trace is the corner’s own normalized trace tr_{r_n} rather than the restriction of tr_{d_n} ; and $\|1 - (-1)\|_{2, \text{tr}_{r_n}} = 2$ for every value of r_n/d_n .

Theorem 4.1 (central involution obstruction). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for some $a \in \Gamma$. If $w := u^2$ is a nontrivial central involution of H , then every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ maps w to the identity, and H is not MF.

Proof. Let $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism and suppose $\Theta(w) \neq 1$.

The (-1) -spectral corner. Choose coordinate unitary lifts $(V_{g,n})$ of Θ (Lemma 2.2) and put $p_n = \frac{1}{2}(1 + V_{w,n})$, the averaging operator of the two-element subgroup $F = \{1, w\}$. Because the lifts are asymptotically multiplicative and $w^2 = 1$, the Hermitian part of p_n is asymptotically idempotent, so its spectrum clusters at $\{0, 1\}$ and rounding at $\frac{1}{2}$ by continuous functional calculus produces genuine projections at operator-norm distance $o(1)$ from p_n ; centrality of w makes them asymptotically central for all of $\Theta(H)$. Let q_n be their complements. If q_n vanished for all large n the averages would converge to 1 and $\Theta(w)$ would be 1; so $q_n \neq 0$ along infinitely many coordinates, which we retain and relabel by $\mathbb{N}$. Compressing every lift to $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ and polar-correcting the compressions — each is almost unitary in the corner, by asymptotic centrality — gives an operator-norm asymptotic representation $(W_{g,n})$ of H on nonzero blocks with

$$\|W_{w,n} + 1\| \rightarrow 0.$$

Equip each corner block with its own normalized trace tr_{r_n} ; all estimates below are relative to r_n and independent of the ratio r_n/d_n .

Transport on the corner. Since c centralizes $\iota(\Gamma)$, the sequence $(W_{c,n})$ centralizes $(W_{\iota(\gamma),n})$ asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. By Theorem 3.4 applied to $x_n = W_{c,n}$, the conjugated sequence $(W_{d,n})$ lies in the same asymptotic commutant. Therefore $\|W_{u,n} - 1\|_2 \rightarrow 0$, and squaring gives $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$.

The contradiction. The relation $w = u^2$ gives $\|W_{u,n}^2 - W_{w,n}\| \rightarrow 0$, while $W_{w,n} \rightarrow -1$ in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence $\|W_{u,n}^2 + 1\|_2 \rightarrow 0$ and $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$, so $\|2 \cdot 1\|_2 \rightarrow 0$.

0, which is false. Thus $\Theta(w) = 1$. Since $w \neq 1$, no homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ is injective, and H is not MF. $\square$

The same conclusion holds for an arbitrary finite normal subgroup of N_{conj} (Theorem S6.3), and for an arbitrary normal property-(T) subgroup $K \leq N_{\text{conj}}$, with a Kazhdan spectral projection in place of the two-element average and with neither centrality nor torsion assumed (Theorem S6.6).

5. THE KAZHDAN–CLIFFORD CONSTRUCTION

Let Γ be a finitely presented Kazhdan group, let α be a proper injective endomorphism of it, and let $a \in \Gamma \setminus \alpha(\Gamma)$.

With $d = tct^{-1}$ and $u = [d, a]$, put

$$w = u^2.$$

Since d is an involution, $w = [d, ada^{-1}]$, a word in the generators, so centrality of w is a relation among them; centrality then forces $w^2 = 1$ by (5) below.

Theorem 5.1 (Kazhdan–Clifford construction). *Let Γ be a finitely presented group with property (T), let $\alpha: \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

and impose centrality of w . More explicitly, for a finite generating set S_Γ it is enough to impose the conjugation and commutation relations for $s \in S_\Gamma$, together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

Thus w commutes with a generating set of the entire resulting group. The group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \rightarrow E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism

$$E(\Gamma, \alpha, a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

to $\mathcal{U}(\mathcal{Q})$ maps w to 1. Consequently $E(\Gamma, \alpha, a)$ is not MF.

Proof. Finite presentability follows from a relative finite presentation: start with the free product of Γ and the free group on t, c , and quotient by the finitely many displayed relations for a finite generating set of Γ .

The relation $w^2 = 1$ is redundant. Put $b = ada^{-1}$. Both d and b are involutions, $w = [d, b] = (db)^2$, and direct cancellation gives

$$(5) \quad dwd^{-1} = w^{-1}.$$

Centrality also gives $dwd^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

concretely. Let Γ_∞ be the direct limit of the system $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$, whose n th copy of Γ we call its n th level; injectivity of α passes to the limit, so the level-zero map embeds Γ in Γ_∞ . Applying α levelwise induces an injective endomorphism σ of Γ_∞ . It is surjective, hence an automorphism, because an element at level k equals σ of the same element at level $k + 1$. Put $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$, with stable letter t implementing σ ; then $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$. Write $X = G/\Gamma$ for the coset space of the level-zero copy. Let G permute the Clifford generators $(e_x)_{x \in X}$ and map c to e_Γ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows both that $w \neq 1$ and that the canonical map of Γ is injective.

Finally, $w = u^2$ is a nontrivial central involution. By the central involution obstruction (Theorem 4.1), every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps w to the identity. Since $w \neq 1$, none is faithful. $\square$

Remark 5.2 (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections S1–S2 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group P contains a copy of every finitely presented torsion-free group, in particular a direct product $P_1 \times P_2$ with $P_i \cong P$. Then $\alpha: P \xrightarrow{\cong} P_1 \leq P$, together with any $a \in P_2 \setminus \{1\}$, satisfies the hypotheses of Theorem 5.1. The group $E(P, \alpha, a)$ contains involutions whatever the base.

6. A FINITELY PRESENTED NON-MF GROUP

Specialize Theorem 5.1 to

$$(6) \quad \Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I).$$

The affine group Γ is finitely presented and has property (T) [BHV, Example 1.7.4(i)]; Section S1 also proves this from the presentation used below, by an exact rational sum-of-squares certificate. The endomorphism α is injective because doubling is injective on $\mathbb{Z}^3$, it is proper because $e_1 \notin 2\mathbb{Z}^3$, and $a \notin \alpha(\Gamma)$ for the same reason. Theorem 5.1 therefore applies, and relative to Γ the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \left| \begin{array}{ll} t\gamma t^{-1} = \alpha(\gamma), & [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, & [w, g] = 1 \end{array} \right. \right\rangle,$$

where S_Γ is a finite generating set of Γ , g runs over the generators, and $w = [tct^{-1}, a(tct^{-1})a^{-1}]$. Expanding a standard presentation of the affine group in the generators v_1, v_2, v_3 for the translations and x, y, z for the linear

part turns this into the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$(7) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *base*

$$(8) \quad \mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (7),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write $\iota: \mathcal{B} \rightarrow E$ for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

Definition 6.1. Use the six base generators and the relations displayed in (8). Define

$$(9) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \mid \begin{array}{l} \text{(B) all relations in (8),} \\ \text{(T) } tv_it^{-1} = v_i^2 \ (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C) } c^2 = 1, \\ [c, g] = 1 \ (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W) } [w, g] = 1 \ (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$. This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Family (B) is a presentation of the base; family (T) conjugates each base generator by t into the base; family (C) makes c an involution commuting with the base generators; and family (W) makes w central. Centrality together with $c^2 = 1$ implies $w^2 = 1$ by (5). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

A Tietze reduction removes two generators and nine relators (Section S1).

Since $d = tct^{-1}$ is an involution, the distinguished word

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}]$$

is the square $[d, \iota(v_1)]^2$.

Lemma 6.2 (the distinguished word is a square). *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

Proof of Theorem A. That $w \neq 1$ is Proposition S2.3.

Let (d_n) be given, write $\mathcal{Q} = \mathcal{Q}((d_n))$, and let $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism. Apply the central involution obstruction (Theorem 4.1) with $\Gamma = \mathcal{B}$, the canonical homomorphism ι , the elements $t, c \in E$, and $a = v_1$. Its hypotheses are the displayed relators: Proposition S1.1 gives property (T) of $\mathcal{B}$; family (T) gives $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$; and family (C) gives $[c, \iota(\mathcal{B})] = 1$. In E we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 6.2 with $a_1 = \iota(v_1)$ gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 = u^2, \quad u = [d, \iota(v_1)].$$

Centrality of w is a defining relator; with $c^2 = 1$ it gives $w^2 = 1$ by (5); and $w \neq 1$ as above. Hence $\Theta(w) = 1$.

An MF embedding of E would be an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, and every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps the nontrivial element w to 1; so E is not MF. Suppose $C_r^*(E)$ were an MF algebra, with an injective, possibly nonunital $*$ -homomorphism $e: C_r^*(E) \rightarrow \mathcal{Q}$. Complement correction gives the injective unitary homomorphism $u \mapsto e(u) + (1 - e(1))$; composing it with the canonical map $E \rightarrow \mathcal{U}(C_r^*(E))$, which is injective because $g \mapsto \lambda_g$ is, exhibits an injective homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$. The canonical map $E \rightarrow \mathcal{U}(C_{\max}^*(E))$ is injective as well, since the left regular representation factors through $C_{\max}^*(E)$ and sends u_g to λ_g , so an MF embedding of $C_{\max}^*(E)$ produces an injective homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$ after the same conversion. $\square$

Corollary 6.3 (uniform finite operator-norm obstruction). *There exist $\delta > 0$ and a finite set $F_0 \subset E$ such that, in every matrix dimension, any unitary-valued map ϕ satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \quad (g, h \in F_0)$$

also satisfies $\|\phi(w) - 1\| < 1$.

Proof. Fix an exhaustion $F_1 \subseteq F_2 \subseteq \dots$ of E by finite sets. If no such pair (δ, F_0) exists, then for each n there is a unitary-valued ϕ_n with multiplicative defect at most $1/n$ on F_n and $\|\phi_n(w) - 1\| \geq 1$. The sequence (ϕ_n) is an operator-norm asymptotic representation of E , so by Lemma 2.2 its class is a homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$, and $\|\Theta(w) - 1\| \geq 1$ gives $\Theta(w) \neq 1$, against Theorem A. $\square$

The argument is a diagonal one and bounds neither δ nor $|F_0|$; whether an effective modulus follows from the rational spectral certificate of Section S1 is open.

7. CONSEQUENCES OF THE EXAMPLE

Throughout, $d = tct^{-1}$ and $u = [d, \iota(v_1)]$, so that $w = u^2$. Write $\text{Res}_{\text{MF}}(G)$ for the *MF residual* of a countable group G : the intersection of the kernels of all its homomorphisms to $\mathcal{U}(\mathcal{Q})$ (Definition S10.1). Thus G is MF exactly when $\text{Res}_{\text{MF}}(G) = 1$ (Proposition S10.3), and Theorem A says $w \in \text{Res}_{\text{MF}}(E)$ with $w \neq 1$.

The quotient by the central involution.

Theorem B (the quotient of E by $\langle w \rangle$). *The quotient $E/\langle w \rangle$ is not MF. The image of $[\iota(v_1), d]$, the inverse of the distinguished commutator u , is nontrivial there and lies in $\text{Res}_{\text{MF}}(E/\langle w \rangle)$; consequently $\{1, w\}$ is a proper subgroup of $\text{Res}_{\text{MF}}(E)$.*

The proof is in Section S11. If an involution is centralized by the conjugate copy of a Kazhdan subgroup, and its conjugates under that subgroup commute with one another, then every homomorphism to $\mathcal{U}(\mathcal{Q})$ identifies the entire conjugation orbit (Theorem S11.2, proved in Section S11). In E the pairwise commutators of those conjugates of d lie in $\langle w \rangle$, so in $E/\langle w \rangle$ the element d satisfies these hypotheses, and every homomorphism $E/\langle w \rangle \rightarrow \mathcal{U}(\mathcal{Q})$ maps u to the identity. Thus $E/\langle w \rangle$ is a second finitely presented non-MF group, and Theorem A does not determine $\text{Res}_{\text{MF}}(E)$ (Section S14).

A stably finite non-MF group algebra. Every MF algebra is stably finite [BK]; the converse fails. The canonical trace on $C_r^*(E)$ is faithful, so $C_r^*(E)$ is stably finite (Lemma S12.1), independently of the MF question. Abstract examples follow from the failure of the Connes embedding problem [FGH]; the reduced algebra of the finitely presented group E is an explicit one (Theorem C). For $C_r^*(\text{SL}_3(\mathbb{Z}))$ and $C_r^*(\text{SL}_4(\mathbb{Z}))$ the MF property is undecided; only the stronger PMF property is known to fail, for $\text{SL}_4(\mathbb{Z})$ [Mdis].

Theorem C (the reduced group algebra of E). *$C_r^*(E)$ is a unital separable C^* -algebra which has a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

Proof. $C_r^*(E)$ is unital and separable because E is countable, and by Lemma S12.1 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A. $\square$

The algebra is not nuclear: the canonical map $\mathcal{B} \rightarrow E$ is injective (Section S2), so E contains an infinite Kazhdan group and is nonamenable, whence $C_r^*(E)$ is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear C^* -algebra is quasi-diagonal — therefore remains open.

A sofic example. Let $X = G/\Gamma$ be the coset space of the ascending HNN extension $G = \Gamma_\infty \rtimes \mathbb{Z}$ used in the proof of Theorem 5.1, at the affine data (6), and let

$$W = \text{Cl}(X) \rtimes G$$

be the Clifford group over it. The homomorphism of that proof carrying w to -1 has target W ; Section S2 realizes W over $\mathbb{Q}$, writing Σ for G in the explicit rational model.

Theorem D (a sofic non-MF group). *The Clifford quotient $W = \text{Cl}(X) \rtimes \Sigma$ of Section S2 is a finitely generated sofic group that is not MF; since sofic groups are hyperlinear, W is likewise a hyperlinear group that is not MF. In particular, neither soficity nor hyperlinearity implies the MF property.*

The proof is in Section S12. The image of the doubling map has finite index, so every level of the direct limit has finite orbits on X , every finite subset of $\text{Cl}(X) \rtimes \Gamma_\infty$ lies in a finite-by-sofic subgroup, and the remaining $\mathbb{Z}$ -extension is sofic by Elek–Szabó [ES06]. The surjection $E \rightarrow W$ sends w to the nontrivial Clifford sign ζ , and $w \in \text{Res}_{\text{MF}}(E)$, so functoriality (Lemma S10.2) gives $\zeta \in \text{Res}_{\text{MF}}(W)$ and W is not MF. The extension $1 \rightarrow \text{Cl}(X) \rightarrow W \rightarrow \Sigma \rightarrow 1$ has locally finite kernel and residually finite quotient, hence MF kernel and MF quotient, while W is not MF: the MF property is not closed under extensions (Section S12). Corollary S10.10 gives the corresponding failure for quotients, through $F_8 \twoheadrightarrow E$.

Theorem E (a finitely presented sofic non-MF group). *The group E of Definition 6.1 is sofic, hence hyperlinear. Consequently E is itself a finitely presented sofic group that is not MF, and a finitely presented hyperlinear group that is not MF. It is also sofic and not LEF, and in particular not residually finite.*

The proof is in Section S12.3, through the decomposition $E \cong C(\mathcal{G}) \rtimes \Sigma$ of Proposition S12.2: here $\mathcal{G}$ is the graph on X whose cliques are the blocks of the coset space, and W is the quotient of E in which lamps at sites of different blocks anticommute.

Supplementary material

The supplement has four parts. Sections S1–S4 verify the explicit group and the two theorems the main text applies to it. Sections S5–S8 develop the transport argument in the generality it has, and show that property (T) cannot be dropped. Sections S9–S13 draw the consequences. Sections S14–S16 describe what the method does not decide, and give two auxiliary constructions.

S1. THE BASE, AND A SMALLER PRESENTATION

The base $\mathcal{B}$ of (8) is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ (Remark S16.2), for which property (T) is classical [BHV, Example 1.7.4(i)]. It is proved below from the displayed relators, by an exact rational spectral certificate for an auxiliary Steinberg-type presentation, an explicit surjection onto the linear presentation, and a passage from the linear subgroup to the full base that produces an invariant vector from an almost-invariant one.

Six generators suffice. The group E is generated by the six elements v_1, x, y, z, t, c , and has a Tietze-equivalent presentation on them with thirty-two relators. In (9), eliminate $v_3 = xv_1x^{-1}$ and $v_2 = x^2v_1x^{-2}$ by Tietze substitutions and delete the two defining relations; the remaining x -action relation follows from $x^3 = 1$, and the stable-letter, c -commutation, and centrality relations for v_2, v_3 follow from those for v_1 together with $[t, x] = 1$ and $x^3 = 1$. Deleting these nine leaves thirty-two relators on v_1, x, y, z, t, c .

Write $d = tct^{-1}$. The displayed relations give $[d, t\iota(\mathcal{B})t^{-1}] = 1$ and leave $[d, \iota(v_1)d\iota(v_1)^{-1}]$ unconstrained (Figure 1).

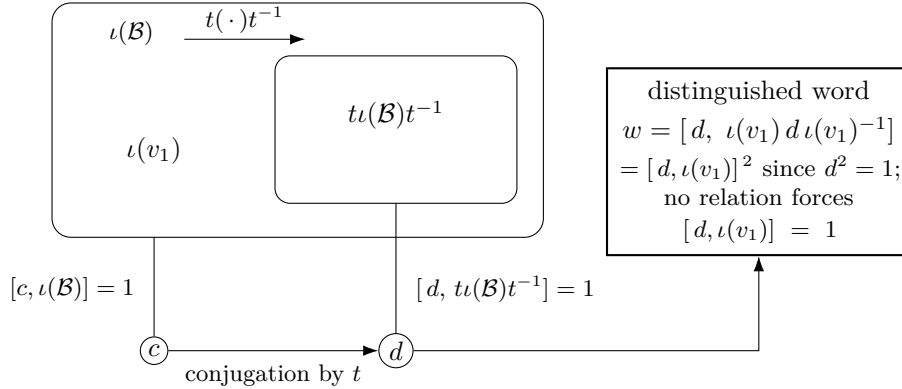


Figure 1: The one-sided conjugation configuration. The relations of (9) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, hence $[d, t\iota(\mathcal{B})t^{-1}] = 1$ for $d = tct^{-1}$; they do not give $[d, \iota(v_1)] = 1$, and $w = [d, \iota(v_1)d\iota(v_1)^{-1}] = [d, \iota(v_1)]^2$. Section S2 proves that the inclusion is strict and that $\iota(v_1)$ lies outside the conjugate copy, as drawn; Theorem 4.1 uses only the inclusion.

S1.1. Property (T) of the base.

Proposition S1.1. *The twenty-relator group $\mathcal{B}$ has property (T), in both the real-orthogonal and complex-unitary formulations.*

Proof. *Step 1: a certified Steinberg-type presentation.* Let P_{13} be the group on the six generators $e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}$ with the thirteen relators

$$\begin{aligned} [e_{12}, e_{23}] &= e_{13}, & [e_{13}, e_{32}] &= e_{12}, & [e_{21}, e_{13}] &= e_{23}, \\ [e_{23}, e_{31}] &= e_{21}, & [e_{31}, e_{12}] &= e_{32}, & [e_{32}, e_{21}] &= e_{31}, \\ [e_{12}, e_{13}] &= [e_{12}, e_{32}] = [e_{13}, e_{23}] = [e_{21}, e_{23}] = [e_{21}, e_{31}] = [e_{31}, e_{32}] = 1, \\ (e_{12} e_{21}^{-1} e_{12})^4 &= 1, \end{aligned}$$

the elementary-matrix relations in rank three. The degree-one Hodge Laplacian Δ_1 of this presentation, a matrix over the real group ring of P_{13} , admits the exact decomposition

$$\Delta_1 = \frac{1}{250} I + Q^* Q + R,$$

in which Q and R are matrices over the group ring with rational coefficients supported on finitely many group elements, and every row and every column of R has coefficient ℓ^1 norm at most $\frac{1}{500}$. Clearing denominators makes both sides equal integer combinations of finitely many group elements, so the decomposition and the two bounds are finite identities and inequalities over $\mathbb{Z}$, evaluated in exact arithmetic; the certificate data are in the supplementary material. Together they bound Δ_1 below by $\frac{1}{500}$ in every orthogonal representation. Testing that bound on coboundaries gives the quadratic gap

$$\|\Delta\xi\|^2 \geq \frac{1}{500} \langle \Delta\xi, \xi \rangle$$

for the six-generator Laplacian Δ , so the spectrum of Δ lies in $\{0\} \cup [\frac{1}{500}, \infty)$. The kernel of Δ is the space of invariant vectors, so in a representation with no nonzero invariant vector that kernel is trivial and Δ is bounded below. Hence a unit vector almost fixed by the six generators has small Laplacian energy and is close to an invariant vector. This yields a Kazhdan pair: P_{13} has property (T). Certificates of this kind go back to Ozawa's algebraic characterization of property (T) [Oz] and its computer-assisted implementations for $\mathrm{SL}_3(\mathbb{Z})$ [NT].

Step 2: the explicit surjection onto the linear presentation. Map the six generators to the words

$$\begin{aligned} e_{12} &\mapsto y^{-1} x z y^{-1} x^{-1}, & e_{13} &\mapsto x z y^{-1} x^{-1} y, & e_{21} &\mapsto y^{-1} x y x z, \\ e_{23} &\mapsto x^{-1} y^{-1} x z y^{-1}, & e_{31} &\mapsto x y^{-1} x z y^{-1} x, & e_{32} &\mapsto x^{-1} z y^{-1} x^{-1} y^{-1} x^{-1} \end{aligned}$$

in the linear presentation $\mathcal{R}$ of (7). Each of the thirteen relators of P_{13} reduces to the identity in $\mathcal{R}$ using only the eight relators of (7); the thirteen reductions are finite words and are carried out in the supplementary material. The assignment therefore defines a homomorphism $P_{13} \rightarrow \mathcal{R}$, and the same reductions exhibit explicit words in the six images whose values are z and

xy . Those two elements generate $\mathcal{R}$: writing $g = z$ and $h = xy$, the eight relators give the generating-pair word of [CLV]

$$y = (ghgh^{-1}g^{-1}hg^{-1}h^{-1}g^{-1}h^{-1}g^{-1}h)^2,$$

and then $x = hy^{-1}$. Hence the homomorphism is surjective. Property (T) passes to quotients, so $\mathcal{R}$ has property (T).

Step 3: from an almost-invariant vector to an invariant one over the base. The generators x, y, z of $\mathcal{B}$ satisfy the relators of $\mathcal{R}$, so $\mathcal{R}$ maps to the linear subgroup of $\mathcal{B}$ they generate, and Step 2 gives a finite set S of words in x, y, z and a $\kappa > 0$ forming a Kazhdan pair for it. A Kazhdan pair remains one when its constant is decreased, so assume $\kappa \leq 1$. Let ρ be an orthogonal representation of $\mathcal{B}$ with a unit vector ξ displaced by less than $\kappa/64$ by every element of the finite control set consisting of S together with the translations v_2, v_3 . Since $v_1 = xv_2x^{-1}$ by the displayed x -action, any vector η fixed by the linear subgroup satisfies $\|\rho(v_1)\eta - \eta\| = \|\rho(v_2)\eta - \eta\|$, so v_1 needs no separate control. We construct a nonzero invariant vector near ξ as follows.

- (1) Projecting ξ to the subspace of vectors fixed by the linear subgroup gives such a fixed vector p with $\|p - \xi\| < 1/64$, by the standard estimate that this projection moves a unit vector by at most the ratio of its displacement under the Kazhdan set to the Kazhdan constant.
- (2) Translations move p by a bound independent of the lattice vector. Three explicit words in x, y, z act on the translation lattice as elementary transvections. Consequently every lattice element is a product of two conjugates, by elements of the linear subgroup, of basis translations. The three words and the elimination that produces this factorization are finite data of the same kind as the thirteen reductions of Step 2, and are in the supplementary material. A conjugate rvr^{-1} with r in the linear subgroup displaces the fixed vector p exactly as much as v does. Each control translation displaces p by less than $1/64 + \kappa/64 + 1/64 = (\kappa + 2)/64 \leq 3/64$, its displacement bound at ξ transferring across $\|p - \xi\|$ at both ends. Hence every translation displaces p by at most $1/8$.
- (3) The orbit of p under the translation subgroup is bounded, so the Hilbert-space circumcenter argument gives a translation-fixed vector q with $\|q - p\| \leq 1/8$; the orthogonal projection η of p to the translation-fixed subspace is the nearest fixed vector, so $\|\eta - p\| \leq \|q - p\| \leq 1/8$. The linear subgroup normalizes the translation subgroup and fixes p , so η is fixed by it as well. Translations and the linear generators generate $\mathcal{B}$, so η is $\mathcal{B}$ -invariant. Finally, $\|\eta - \xi\| \leq 1/8 + 1/64 < 1$ keeps η nonzero.

Thus almost-invariant vectors produce invariant vectors and $\mathcal{B}$ has property (T) in the real-orthogonal formulation; realification and complexification identify it with the complex-unitary formulation. $\square$

$P_{13} \cong \mathrm{SL}_3(\mathbb{Z})$, and $\mathcal{B}$ is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ (Section S16).

S2. NONTRIVIALITY OF w

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2].

Construction S2.1 (Clifford groups). For a set X , use the real Clifford-algebra convention generated by $(c_x)_{x \in X}$ with $c_x^2 = 1$ and $c_x c_y = -c_y c_x$ for $x \neq y$, and let $\text{Cl}(X)$ be the subgroup of units generated by -1 and the c_x . Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Thus $\text{Cl}(X)$ is a central extension, by the sign ζ , of the elementary abelian 2-group on the points of X : the commutator of two distinct generators is ζ , unlike the commuting generators of a classical lamplighter. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ .

Lemma S2.2 (linear model). *For $M \in \text{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

Let $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \text{diag}(2, 2, 2, 1)$. Direct matrix computation proves that the six matrices satisfy all twenty relators of $\mathcal{B}$ and that

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z . Moreover the matrix assigned to v_1 does not lie in $\text{range}(\bar{\alpha})$.

Proof. The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\text{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$. However $D^{-1}v_1D$ has entry $1/2$ in position $(1, 4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. $\square$

Form the direct limit $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$, the direct limit of $\bar{\alpha}$, extend it by its shift τ , and call the resulting group $\Sigma = T \rtimes \mathbb{Z}$. Let $j: \bar{\Gamma} \hookrightarrow \Sigma$ be the level-zero map. Put $X = \Sigma/j(\bar{\Gamma})$, with base point $o = j(\bar{\Gamma})$. Since $D^{-1}\bar{v}_1 D \notin \bar{\Gamma}$ (Lemma S2.2), we have $\tau o \neq j(\bar{v}_1)\tau o$, and the Clifford generators at these two points anticommute.

Proposition S2.3. $w \neq 1$ in E .

Proof. Form $W := \text{Cl}(X) \rtimes \Sigma$ as in Construction S2.1. Define a map on the generators of (9) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, \quad c \mapsto c_o. \end{aligned}$$

The exact checks in Lemma S2.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o , so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma S2.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W , the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E . $\square$

When X is infinite so is $\text{Cl}(X)$, but each of its elements involves only finitely many Clifford generators, and every element of Σ used above is a rational 4×4 matrix. The two points are distinct because $D^{-1}\bar{v}_1 D$ has the entry $1/2$ in position $(1, 4)$ and therefore does not lie in $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$.

S3. THE MF EQUIVALENCES

An *approximate unitary representation* for a countable group G , a finite set $F \subseteq G$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$ is a map $V: G \rightarrow \mathcal{U}(r)$, for some positive dimension r , such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\leq \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\geq \delta \quad (g \neq h \in F). \end{aligned}$$

The local formulation in Proposition 2.3 asks for such a model with $\delta = 1$ for every finite F and every $\varepsilon > 0$.

Proof of Proposition 2.3. Lemma 2.2 identifies $\prod_n \mathcal{U}(d_n)/\mathcal{N}$ with $\mathcal{U}(\mathcal{Q})$, which is the first equivalence once the dimension convention is normalized.

Dimension normalization. Given positive dimensions d_n , replace the n th sequence of unitary lifts by

$$\tilde{V}_n(g) = I_{\mathbb{C}^{d_0}} \oplus \cdots \oplus I_{\mathbb{C}^{d_{n-1}}} \oplus V_n(g)$$

on $\mathbb{C}^{d_0} \oplus \cdots \oplus \mathbb{C}^{d_n}$, of strictly increasing dimension $D_n = d_0 + \cdots + d_n$. Block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements, so for all g, h

$$\begin{aligned} \|\tilde{V}_n(g)\tilde{V}_n(h) - \tilde{V}_n(gh)\| &= \|V_n(g)V_n(h) - V_n(gh)\|, \\ \|\tilde{V}_n(g) - \tilde{V}_n(h)\| &= \|V_n(g) - V_n(h)\|. \end{aligned}$$

Both the class in the quotient and its injectivity are therefore unchanged, and the converse is immediate.

Suppose first that $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$ is injective, and choose unitary lifts $u_n(g)$ of $\rho(g)$. Fix a finite set $F \subseteq G$ and $\varepsilon > 0$. For each ordered pair $g \neq h$ in F , injectivity says that $(u_n(g)^{-1}u_n(h))_n$ does not converge to the identity in operator norm. Hence there are $\delta_{g,h} > 0$ and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer $N_{g,h}$ with $8 < N_{g,h} \delta_{g,h}^2$. The following estimate then applies: if $u, v \in \mathcal{U}(d)$ satisfy $\|u - v\| \geq \delta$, then $\|u^{\otimes p} - v^{\otimes p}\| > 1$ for some $p \leq N$ whenever $8 < N\delta^2$. Indeed, put $z = u^*v$, so that $\|u - v\| = \|1 - z\|$ and z has a spectral value $e^{2\pi i\alpha}$ with $|1 - e^{2\pi i\alpha}| = 2\sin(\pi\alpha') \geq \delta$, where $\alpha' = \min(\{\alpha\}, 1 - \{\alpha\})$; since $(u^{\otimes p})^*v^{\otimes p} = z^{\otimes p}$ has $e^{2\pi ip\alpha}$ in its spectrum,

$$\|u^{\otimes p} - v^{\otimes p}\| \geq |1 - e^{2\pi ip\alpha}| = 2\sin(\pi \min(\{p\alpha\}, 1 - \{p\alpha\})),$$

which exceeds 1 as soon as $\{p\alpha\}$ lies in $(1/6, 5/6)$. The bound $2\sin(\pi\alpha') \geq \delta$ gives $\alpha' \geq \delta/(2\pi)$; consecutive multiples of α move $\{p\alpha\}$ by α' in a fixed direction modulo 1, which is less than the length $2/3$ of that interval, so some $p \leq \lceil 1/(6\alpha') \rceil \leq \pi/(3\delta) + 1 < N$ lands in it. The quotient homomorphism law makes all multiplicative defects for pairs in F arbitrarily small at every sufficiently large coordinate. Choose, for each $g \neq h$, a coordinate with the displayed separation and with every required defect at most $\varepsilon/N_{g,h}$. Then the p th tensor power separates the designated pair by more than 1, while the tensor-product telescoping identity bounds each new defect by $p\varepsilon/N_{g,h} \leq \varepsilon$. Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in F is separated by at least 1 and every required product has defect at most ε . This gives the normalized approximate representation.

Conversely, suppose approximate unitary representations with separation constant 1 exist for all finite sets and tolerances. Choose models along an exhaustion with errors tending to zero. Each fixed pair g, h is then eventually represented multiplicatively, and each fixed pair $g \neq h$ is eventually separated by 1. Hence the classes define an injective homomorphism into $\prod_n \mathcal{U}(r_n)/\mathcal{N}$, with no condition on the dimension sequence. In particular every nonidentity element is separated by some homomorphism to a matrix quotient, so the MF residual of Definition S10.1 is trivial, and the block-sum diagonalization in the proof of Proposition S10.3(2) then assembles the separating sequences

into a single faithful homomorphism to $\mathcal{U}(\mathcal{Q})$; that diagonalization uses the dimension normalization above and nothing else from the present proposition. $\square$

S4. THE FINITE-STAGE PROOF

Theorem 3.4 also has a proof at finite stages, in which every estimate is an explicit inequality in κ , $|S|$, the uniform bound on (x_n) , and the multiplicative defects of U_n , conditional on the spectral-gap estimate (i) below. That estimate is not reducible to a threshold on finitely many defects: property (T) constrains exact unitary representations, while at a finite stage the matrix models generate only a representation of the free group on S . Making it effective as well would need a sum-of-squares certificate for property (T) in the sense of [Oz].

Finite-stage proof of Theorem 3.4. Property (T) provides a finite symmetric generating set $S \subseteq \Gamma$ with $1 \in S$ and a constant $\kappa \in (0, 1]$ such that (S, κ) is a Kazhdan pair. As in the proof of Section 3, each group element acts by conjugation, $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$, on $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, and operator-norm almost multiplicativity of U_n makes $g \mapsto \text{Ad } U_n(g)$ an operator-norm asymptotic representation on these Hilbert spaces.

The almost-fixed spectral subspace. Let H_n be the Hermitian part of the average $\frac{1}{|S|} \sum_{s' \in S} \text{Ad } U_n(\iota(s'))$, fix the threshold

$$\theta = \frac{1}{2} \left(\left(1 - \frac{\kappa^2}{4|S|} \right) + 1 \right) < 1,$$

and let P_n be the spectral projection of H_n for the part of the spectrum above θ .

- (i) *Displacement.* For every $\gamma \in \Gamma$, $\|(\text{Ad } U_n(\iota(\gamma)) - 1) P_n\| \rightarrow 0$. This comes from a quantitative spectral-gap estimate: for every pair a, b with $1 - \kappa^2/(4|S|) < a$ and $b < 1$, once the multiplicative defects of U_n on the finitely many group elements involved are small enough, no eigenvalue of H_n lies in $[a, b]$. Indeed, a unit eigenvector η with $H_n \eta = \lambda \eta$ satisfies

$$\frac{1}{|S|} \sum_{s' \in S} \|(\text{Ad } U_n(\iota(s')) - 1) \eta\|_2^2 = 2(1 - \lambda),$$

so $\lambda \geq a > 1 - \kappa^2/(4|S|)$ forces $\|(\text{Ad } U_n(\iota(s')) - 1) \eta\|_2^2 < \kappa^2/2$ for every $s' \in S$; by the Kazhdan inequality η is then close to an invariant vector, on which H_n is within the multiplicative defects of 1, against $\lambda \leq b < 1$. Hence the spectrum of H_n above θ is eventually within any prescribed δ of 1, so a unit vector $\eta \in \text{ran } P_n$ has $\langle (1 - H_n) \eta, \eta \rangle \leq \delta$; since $1 - H_n$ is the average of the positive operators $1 - \text{Re } \text{Ad } U_n(\iota(s'))$, each generator satisfies $\frac{1}{2} \|(\text{Ad } U_n(\iota(s')) - 1) \eta\|_2^2 \leq |S| \delta$, and a word in S of length ℓ moves η by at most $\ell \sqrt{2|S|} \delta$ plus the accumulated multiplicative defects.

- (ii) *Spectral concentration.* Write $\xi_n \in L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ for the vector x_n . The commutator hypothesis says each $\text{Ad } U_n(\iota(s'))$ almost fixes ξ_n , so the residual $(1 - H_n)\xi_n$ is Hilbert–Schmidt null. On $\text{ran}(1 - P_n)$ the operator $1 - H_n$ is bounded below by $1 - \theta$, and it commutes with P_n , so every vector η satisfies

$$\|(1 - P_n)\eta\|_2 \leq (1 - \theta)^{-1} \|(1 - H_n)\eta\|_2.$$

Applied to $\eta = \xi_n$, this shows ξ_n is asymptotically concentrated in $\text{ran } P_n$.

Reversing the one-sided projection estimate. Put $T_n = \text{Ad } U_n(s)$ and $Q_n = T_n P_n T_n^*$, a projection of the same rank as P_n . Since $\|(1 - Q_n)P_n\| = \|(1 - P_n)T_n^* P_n\|$, it is enough to bound the latter. Let $\eta \in \text{ran } P_n$ be a unit vector and let $s' \in S$. The hypothesis $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$ supplies a single $\delta_{s'} \in \Gamma$ with $s\iota(s')s^{-1} = \iota(\delta_{s'})$, and

$$(\text{Ad } U_n(\iota(s')) - 1)T_n^* \eta = T_n^*(\text{Ad } U_n(\iota(\delta_{s'})) - 1)\eta + R_{n,s'},$$

where $\|R_{n,s'}\|$ is bounded by the multiplicative defects of U_n on $\{s, \iota(s'), s^{-1}\}$ and therefore tends to 0. By (i) so does the first term, so $T_n^* \eta$ is asymptotically fixed by every generator, and by (ii) it lies asymptotically in $\text{ran } P_n$. Hence $\|(1 - Q_n)P_n\| \rightarrow 0$, the finite-stage form of the one-sided inclusion $P \leq VPV^*$. Since $\text{rank } P_n = \text{rank } Q_n$, the equal-rank projection lemma — for projections p, q of equal finite rank, $\|(1 - q)p\| \leq \varepsilon < 1$ implies $\|(1 - p)q\| \leq \varepsilon/\sqrt{1 - \varepsilon^2}$ — reverses the containment:

$$\|(1 - P_n)Q_n\| \rightarrow 0.$$

Combining the estimates. Fix γ and $\varepsilon > 0$, let $M \geq 1$ bound $\|x_n\|$, and put $q = \min(1, \varepsilon/(40(M^2 + 1)))$. For all large n ,

$$\|(1 - P_n)\xi_n\|_2^2 \leq q, \quad \|(1 - P_n)Q_n\| \leq q, \quad \|(A_n - 1)P_n\| \leq q,$$

where $A_n = \text{Ad } U_n(\iota(\gamma))$. Decompose the transported vector along P_n and Q_n :

$$T_n \xi_n = P_n Q_n T_n \xi_n + (1 - P_n)Q_n T_n \xi_n + (1 - Q_n)T_n \xi_n.$$

Since $Q_n = T_n P_n T_n^*$ and T_n is conjugation by a unitary,

$$\|(1 - Q_n)T_n \xi_n\|_2 = \|T_n(1 - P_n)\xi_n\|_2 = \|(1 - P_n)\xi_n\|_2 \leq q^{1/2};$$

the middle term obeys $\|(1 - P_n)Q_n T_n \xi_n\|_2 \leq \|(1 - P_n)Q_n\| \|\xi_n\|_2 \leq qM$, and the displacement estimate moves the first term by at most $q \|\xi_n\|_2 \leq qM$. Applying $A_n - 1$, whose operator norm is at most 2, to the last two terms and using the displacement bound on the first gives

$$\|(A_n - 1)T_n \xi_n\|_2^2 \leq (qM + 2qM + 2q^{1/2})^2 \leq 18q^2 M^2 + 16q \leq \varepsilon.$$

Since $T_n \xi_n$ is the vector of $U_n(s)x_n U_n(s)^*$, this proves the asserted convergence. Once the threshold of (i) is fixed, every bound used above is explicit in κ , $|S|$, M , and the multiplicative defects of U_n on the finitely many group elements involved. $\square$

S5. TRANSPORT AT ARBITRARY WEIGHTS

The transport theorem extends to arbitrary weights ν_n in place of the dimension; Theorem 3.4 is the case $\nu_n = d_n$.

Theorem S5.1 (transport at an arbitrary weight, and of intertwiners). *Let Γ , H , ι , and s be as in Theorem 3.4.*

- (1) *Let (U_n) be as there, let (ν_n) be nonnegative weights, and suppose there is C with $\text{Tr}(x_n^* x_n) \leq C \nu_n$ for all n , and that for every $\gamma \in \Gamma$ and every $\varepsilon > 0$, eventually*

$$\text{Tr} |x_n - U_n(\iota(\gamma)) x_n U_n(\iota(\gamma))^*|^2 \leq \varepsilon \nu_n.$$

Then $U_n(s) x_n U_n(s)^$ and $U_n(s)^* x_n U_n(s)$ satisfy both conditions as well.*

- (2) *Let $U_n^1: H \rightarrow \mathcal{U}(d_n^1)$ and $U_n^2: H \rightarrow \mathcal{U}(d_n^2)$ be operator-norm asymptotic representations and let (x_n) be uniformly operator-norm bounded $d_n^1 \times d_n^2$ matrices with*

$$\frac{\text{Tr} |x_n - U_n^1(\iota(\gamma)) x_n U_n^2(\iota(\gamma))^*|^2}{d_n^1 + d_n^2} \longrightarrow 0 \quad (\gamma \in \Gamma).$$

Then $U_n^1(s) x_n U_n^2(s)^$ satisfies the same convergence for every $\gamma \in \Gamma$.*

Proof. (1) Give each coordinate space with $\nu_n > 0$ the inner product $\langle x, y \rangle = \text{Tr}(y^* x) / \nu_n$; an index with $\nu_n = 0$ forces $x_n = 0$. Conjugation by a unitary is unitary for every weight, and both hypotheses and both conclusions concern only the renormalized norms, so the ultraproduct proof of Theorem 3.4 applies with tr_{d_n} replaced by $\text{Tr}(\cdot) / \nu_n$ and nothing else changed: the bound $\text{Tr}(x_n^* x_n) \leq C \nu_n$ defines the ultraproduct vector, the adjoint actions remain operator-norm almost multiplicative because the weight does not enter the operator norm, and finiteness of the norm ultraproduct turns $P \leq V P V^*$ into $P = V P V^*$, which gives both directions.

(2) The block sum $U_n^\oplus = U_n^1 \oplus U_n^2$ is an operator-norm asymptotic representation, since block-diagonal operator norms are maxima. The off-diagonal embedding of x_n has operator norm $\|x_n\|$, its commutator with $U_n^\oplus(\iota(\gamma))$ is the off-diagonal embedding of the intertwining defect — whose normalized Hilbert–Schmidt norm on the block space has the denominator $d_n^1 + d_n^2$ — and conjugation by $U_n^\oplus(s)$ acts on the embedded corner by $x_n \mapsto U_n^1(s) x_n U_n^2(s)^*$. Theorem 3.4 applied to $U_n^\oplus$ transports the embedded sequence. Replacing the denominator $d_n^1 + d_n^2$ by an arbitrary weight ν_n and applying (1) to the block sum instead of Theorem 3.4 gives the intertwiner transport at every weight. $\square$

Taking $\nu_n = d_n$ gives Theorem 3.4 with the uniform operator-norm bound relaxed to a uniform bound on $\text{Tr}(x_n^* x_n) / d_n$. Taking $\nu_n = k_n$, the rank of Step 3 in the proof of Theorem S11.2, gives the normalization used there.

Invariant tensors of type (p, q) may be identified with Hilbert–Schmidt intertwiners from the q th to the p th tensor power. Tensor powers of unitaries

are exactly multiplicative and satisfy $\|A^{\otimes p} - B^{\otimes p}\| \leq p \|A - B\|$, so tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations, and Theorem S5.1(2) applies to every pair of them: at each fixed tensor type and each weight, a sequence asymptotically invariant for the conjugate copy of the Kazhdan image is already asymptotically invariant for the whole of it.

S6. OBSTRUCTIONS FROM A ONE-SIDED CONJUGATION DATUM

In each result below a subgroup of H lies in the kernel of every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$, because it is at once 2-norm trivial and 2-norm far from the identity on a corner. The hypothesis producing the corner varies:

- Theorem S6.3: a *finite* normal subgroup of N_{conj} , the corner being the complement of its rounded averaging operator — the case of the central involution of Section 4;
- Theorem S6.6: a normal property-(T) subgroup of N_{conj} , the corner being a spectral projection of a Kazhdan average, with neither finiteness, centrality nor torsion assumed;
- Theorem S6.7: the same statement with the conjugation datum discarded and its only consequence, 2-norm triviality, taken as the hypothesis;
- Corollary S6.8: that statement applied to the subgroup $\mathfrak{D}(H, L)$ of (10), which depends on L alone and not on a choice of t and c .

Definition S6.1 (Kazhdan conjugation datum). Let H be a countable group. A *Kazhdan conjugation datum* in H is a tuple (Λ, ι, t, c) , where Λ is a countable group with property (T), $\iota: \Lambda \rightarrow H$ is a group homomorphism, not assumed injective, and $t, c \in H$, subject to

- (M1) $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;
- (M2) $c \iota(\lambda) = \iota(\lambda) c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The datum determines the normal closure

$$N_{\text{conj}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

For a finitely generated Λ , conditions (M1)–(M2) are imposed by finitely many relations once generating sets are fixed. The relations make c commute with $\iota(\Lambda)$ and d commute with the subgroup $t\iota(\Lambda)t^{-1}$, but they do not constrain the commutators $[d, \iota(\lambda)]$; N_{conj} is their normal closure. Every finite normal subgroup of H contained in N_{conj} , and more generally every normal property-(T) subgroup contained in it, lies in the kernel of every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ (Theorems S6.3 and S6.6).

Definition S6.2 (2-norm triviality). Let $U = (U_n)$ be an operator-norm asymptotic representation of a countable group H and let ω be an ultrafilter refining the cofinite filter. Write

$$K_2^\omega(U) = \{g \in H : \lim_\omega \|U_n(g) - 1\|_2 = 0\},$$

the kernel of the homomorphism induced by U into the tracial ultraproduct of the matrix blocks along ω , and hence a normal subgroup of H . An element g is *2-norm trivial* if $g \in K_2^\omega(U)$ for every such U and every such ω . Convergence of the full sequence, $\|U_n(g) - 1\|_2 \rightarrow 0$, gives membership along every ω at once, and is how the condition is verified below.

Theorem S6.3 (finite-normal obstruction criterion). *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c) , and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\text{conj}}$. Then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Proof. *2-norm triviality of N_{conj} .* Let $U_n: H \rightarrow \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H — in particular, coordinate unitary lifts of a homomorphism to $\mathcal{U}(\mathcal{Q})$ — write $U_{g,n} = U_n(g)$, and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \rightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. Since c centralizes $\iota(\Lambda)$, its lifts belong to the asymptotic commutant in normalized Hilbert–Schmidt norm of $\iota(\Lambda)$. By Theorem 3.4 applied to $x_n = U_{c,n}$, the lifts of $d = tct^{-1}$ lie there as well. Therefore $[d, \iota(\lambda)] \in K_2$ for every $\lambda \in \Lambda$; since K_2 is normal, it contains the normal closure of these commutators: $N_{\text{conj}} \subseteq K_2$, and so $F \subseteq K_2$. Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of N_{conj} is 2-norm trivial (Definition S6.2).

The coordinate averaging corner. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F ; choose coordinate unitary lifts $(V_{g,n})$ of Θ and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and F is finite, (p_n) is asymptotically self-adjoint and idempotent, and normality of F makes it asymptotically central for all of $\Theta(H)$. The Hermitian part of p_n is asymptotically idempotent, so its spectrum clusters at $\{0, 1\}$ and rounding at $1/2$ by continuous functional calculus yields genuine projections at operator-norm distance $o(1)$ from p_n ; the rounded projections inherit the asymptotic centrality. Let q_n be their complements. Nontriviality of Θ on F forces $q_n \neq 0$ along infinitely many coordinates — otherwise the averages converge to 1 and $\Theta|_F$ is trivial. Retain exactly those coordinates and relabel them by $\mathbb{N}$. Compress every lift to the corner $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ with $r_n = \text{rank } q_n > 0$, and polar-correct the compressed matrices: asymptotic centrality makes each compression $q_n V_{g,n} q_n$ almost unitary in the corner, so its polar unitary is at operator-norm distance $o(1)$ and almost-multiplicativity is preserved by the correction. The result is an operator-norm asymptotic representation $(W_{g,n})$ of H on the nonzero corner blocks. Since the corner is the complement

of the rounded average, the finite average vanishes there:

$$(*) \quad \left\| \sum_{f \in F} W_{f,n} \right\| \longrightarrow 0 \quad \text{in operator norm.}$$

The contradiction. Since $F \subseteq N_{\text{conj}}$ and $(W_{g,n})$ is itself an operator-norm asymptotic representation of H , the first step applies to it: every $f \in F$ satisfies $\|W_{f,n} - 1\|_2 \rightarrow 0$. Since F is finite, the maximum of these $|F|$ null sequences is itself null, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts $(*)$, because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no homomorphism to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F . $\square$

The same conclusion holds with N_{conj} replaced by any normal subgroup all of whose elements lie in $K_2^\omega(U)$ for every operator-norm asymptotic representation U of H and every ω ; Theorem S6.5 is that statement for $\mathfrak{D}(H, L)$.

The central involution obstruction (Theorem 4.1) is the case $F = \{1, w\}$, where $p_n = \frac{1}{2}(1 + V_{w,n})$ and the complementary corner gives a model in which $w = -1$ exactly.

S6.1. An intrinsic form. For a subgroup $L \leq H$, let $G^+(L)$ be generated by the elements s satisfying $sLs^{-1} \subseteq L$ — the *one-sided conjugators* of L — and put

$$(10) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G^+(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H,$$

which depends only on the pair (H, L) , not on a choice of datum.

Corollary S6.4 (transport along all one-sided conjugators). *Let (r_n) and $V_{g,n} \in \mathcal{U}(r_n)$ be an operator-norm asymptotic representation of H , let Λ have property (T), let $\iota: \Lambda \rightarrow H$ be a homomorphism, and put $L = \iota(\Lambda)$. Suppose that $(x_n \in M_{r_n}(\mathbb{C}))$ is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of L :*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

Then every $g \in G^+(L)$ acts in both directions on that asymptotic commutant. Explicitly, both

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every $\lambda \in \Lambda$.

Proof. For a one-sided conjugator s , Theorem 3.4 gives forward transport, and the equality $P = VPV^*$ established in its proof gives $V^*PV = P$, hence $V^*\text{Fix} = \text{Fix}$ and reverse transport. The elements acting in both directions form a subgroup of H , which therefore contains $G^+(L)$. $\square$

Theorem S6.5 (the subgroup $\mathfrak{D}$ and the MF residual). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \leq \mathfrak{D}(H, L)$, then every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ maps every element of F to the identity.*

Proof. Let $(U_{h,n})$ be coordinate lifts of an operator-norm asymptotic representation of H and put $K_2 = \{h : \|U_{h,n} - 1\|_2 \rightarrow 0\}$, a normal subgroup of H . If $z \in C_H(L)$, then $(U_{z,n})$ lies in the asymptotic commutant of L , and by Corollary S6.4 so does $(U_{gzg^{-1},n})$ for every $g \in G^+(L)$. Hence each generator $[gzg^{-1}, \ell]$, $\ell \in L$, displayed in (10) belongs to K_2 ; since $K_2 \trianglelefteq H$, it contains their normal closure: $\mathfrak{D}(H, L) \leq K_2$. Every element of $\mathfrak{D}(H, L)$ is therefore 2-norm trivial in every operator-norm asymptotic representation of H . The coordinate averaging corner of Theorem S6.3 uses only that triviality, so it applies to any finite normal $F \leq \mathfrak{D}(H, L)$. $\square$

S6.2. The normal-Kazhdan form of the obstruction. A finite group has property (T), and its averaging operator is the projection onto its fixed space. The next theorem replaces that operator by the spectral projection of a Kazhdan average, and assumes neither finiteness, centrality nor torsion.

Theorem S6.6 (normal-Kazhdan obstruction). *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c) , and let $K \trianglelefteq H$ be a normal subgroup with property (T) such that $K \subseteq N_{\text{conj}}$. Then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Finite groups have property (T), so $K = F$ recovers Theorem S6.3. In Theorem S6.6, K may be infinite, noncentral, and torsion-free.

Proof. The containment $N_{\text{conj}} \subseteq K_2$ from the proof of Theorem S6.3 holds for any normal subgroup, so every element of K is 2-norm trivial in every operator-norm asymptotic representation of H .

Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K . Its image $\bar{H} = \Theta(H)$ is countable, hence MF, and Proposition 2.3, applied along an exhaustion of $\bar{H}$, gives an operator-norm asymptotic representation $(V_{g,n})$ in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models bound only operator-norm distances; at growing dimension they give no lower bound on normalized Hilbert–Schmidt distances. The datum and the subgroup pass to their Θ -images with normality and property (T) preserved, and $\bar{K} = \Theta(K)$ is nontrivial and lies in the subgroup N_{conj} determined by the image datum; it suffices to derive a contradiction for $\bar{H}$.

The spectral projection q_n . Fix a finite symmetric generating Kazhdan set $S \subseteq \bar{K}$ with constant ε_0 , at each coordinate let h_n be the Hermitian part of the average of the models of S , and cut the spectrum of h_n at $\theta = 1 - \varepsilon_0^2/(4|S|)$. Let q_n be the spectral projection of h_n for $[-1, \theta]$. On the complementary range, where the spectrum of h_n lies above θ , every model of every element of $\bar{K}$ is close to the identity in operator norm, by

the Kazhdan-pair displacement estimate. If $q_n = 0$ for all large n , every element of $\bar{K}$ would satisfy $\|V_{k,n} - 1\| \rightarrow 0$, contradicting the operator-norm separation of any fixed nontrivial element of $\bar{K}$. So $q_n \neq 0$ along an infinite set of coordinates, which we retain.

A 2-norm separation on the corner. Equip each retained corner block $q_n M_{d_n}(\mathbb{C}) q_n$ with its renormalized trace. By the choice of q_n as the spectral projection of h_n for $[-1, \theta]$, $q_n h_n q_n \leq \theta q_n$, so the average over S of the real parts of the renormalized traces of the compressed models is at most θ . Hence at every retained coordinate some generator in S has compressed model with real renormalized trace at most θ , and therefore normalized Hilbert–Schmidt distance at least $\sqrt{2(1-\theta)} - o(1)$ from the corner identity after polar correction; passing to a subsequence fixes a single such generator s_0 .

Compression. Normality of $\bar{K}$ makes q_n asymptotically commute with the ambient models: $\|[q_n, V_{g,n}]\| \rightarrow 0$ for every g . Because the models are only asymptotically multiplicative, $V_{g,n} h_n V_{g,n}^*$ is not the Hermitian average over the conjugate generating set $g S g^{-1}$ of $\bar{K}$ but converges to it in operator norm; that set has the same Kazhdan constant. The displacement and concentration estimates of Section S4 apply to either average symmetrically: a vector on which one average has spectrum above θ is almost fixed by each generator of the other set (displacement, transported across the finitely many conjugations by almost-multiplicativity), hence almost lies in the corresponding range for the other average (the concentration inequality). Two-sided almost-containment of the two subspaces implies $\|q_n - V_{g,n} q_n V_{g,n}^*\| \rightarrow 0$, that is, $\|[q_n, V_{g,n}]\| \rightarrow 0$ for every $g \in \bar{H}$. Compressing every model to the corner $q_n M_{d_n}(\mathbb{C}) q_n$ and polar-correcting yields an operator-norm asymptotic representation of all of $\bar{H}$ on nonzero blocks, in which s_0 keeps a definite normalized Hilbert–Schmidt distance from the identity.

The contradiction. The corner models form an operator-norm asymptotic representation of $\bar{H}$, and $\bar{K}$ lies in the subgroup N_{conj} determined by the image datum, so 2-norm triviality applies to them: every element of $\bar{K}$ — in particular s_0 — satisfies $\|W_{s_0,n} - 1\|_2 \rightarrow 0$ on the corner, contradicting the separation just obtained. $\square$

Theorem S6.7 (obstruction from 2-norm triviality). *Let H be a countable group and let $D \leq H$ be a subgroup every element of which is 2-norm trivial (Definition S6.2). Let $K \trianglelefteq H$ be a normal subgroup with property (T) such that $K \leq D$. Then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Proof. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K . The image $\bar{H} = \Theta(H)$ is countable and MF, and Proposition 2.3 provides an operator-norm asymptotic representation of $\bar{H}$ separating every fixed nontrivial element. Every operator-norm asymptotic representation of $\bar{H}$ composes with Θ to one of H , so every element of $\Theta(D)$ is again 2-norm trivial,

while $\bar{K} = \Theta(K) \leq \Theta(D)$ is normal, Kazhdan and nontrivial. The spectral projection q_n , the 2-norm separation at some $s_0 \in \bar{K}$, and the compression constructed in the proof of Theorem S6.6 use only that $\bar{K}$ is normal, Kazhdan and nontrivial, and 2-norm triviality of s_0 on the corner contradicts that separation. $\square$

Applied to the intrinsic defect, Theorem S6.7 gives the companion of Theorem S6.5 for a normal property-(T) subgroup:

Corollary S6.8 (normal-Kazhdan form of the intrinsic residual). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $K \trianglelefteq H$ is a normal subgroup with property (T) and $K \leq \mathfrak{D}(H, L)$, then every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ maps every element of K to the identity.*

Proof. By the proof of Theorem S6.5, every element of $\mathfrak{D}(H, L)$ is 2-norm trivial in every operator-norm asymptotic representation of H . Theorem S6.7 with $D = \mathfrak{D}(H, L)$ concludes. $\square$

The subgroup generated by *all* normal property-(T) subgroups of $\mathfrak{D}(H, L)$ therefore also lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$.

S7. THE FINITE-DIMENSIONAL OBSTRUCTION

Every finite-dimensional representation of E over every field maps w to the identity (Theorem 3.3 with $H = E$). So finite-dimensional linear representations do not separate the points of E , no finite quotient witnesses $w \neq 1$, and the Clifford quotient of Section S2 has to be infinite. The central involution w lies in the kernel of every homomorphism to a finite group, so E is a group of *Deligne type* in the terminology of [BDL, Definition 1.2]. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and E is such a counterexample for the operator norm.

The same calculation proves the final, subgroup-valued assertion of Theorem 3.3. The elements of H whose conjugation action preserves C in both directions form a subgroup. Every one-sided conjugator lies in it by the argument above, hence so does all of $G^+(\Gamma)$. Thus, for $g \in G^+(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence every displayed generator in (10) lies in $\ker \pi$; since $\ker \pi \trianglelefteq H$, the whole normal closure $\mathfrak{D}(H, \Gamma)$ does as well.

Corollary S7.1. *The group E of Definition 6.1 admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, every finite-dimensional linear representation maps w to the identity, and every homomorphism to a finite group maps w to the identity.*

Proof. Apply Theorem 3.3 with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1 ; the hypotheses hold by the defining relations (9). Every

finite-dimensional representation maps w to the identity, and $w \neq 1$ by Proposition S2.3. For a finite target, compose with its faithful left regular representation on the group algebra over $\mathbb{Q}$. The composite is a finite-dimensional linear representation, so it maps w to the identity. Evaluating the resulting identity matrix at the basis vector of the identity element returns the image of w in the finite group. $\square$

Remark S7.2 (a conjugation-invariant size forbids strict compression). The proof of Theorem 3.3 used one feature of finite dimensionality: if $V \subseteq W$ and $\dim V = \dim W < \infty$, then $V = W$. Abstractly, let a conjugation-invariant size σ with values in $\mathbb{N}$ be defined on the subgroups of a group Q , and suppose $K \leq L$ with $\sigma(L) \leq \sigma(K)$ implies $K = L$. If $gKg^{-1} \subseteq K$ then the two have equal size, so the inclusion is an equality: in such a target no subgroup is properly conjugated into itself. Cardinality on the subgroups of a finite group qualifies, which is the finite-quotient argument of Section 1.

The finite-dimensional case is *not* that principle applied to subgroups. The natural candidate size, the dimension $\dim_k \text{End}_H(V)$ of the commutant, is order-reversing rather than order-preserving, and it does not separate nested subgroups: $A_4 < S_4$ in the standard three-dimensional representation have the same commutant $k \cdot 1$. Theorem 3.3 applies the principle to the commutant instead, where the inclusion $C \subseteq uCu^{-1}$ between subspaces of equal finite dimension is an equality (4); it is that equality, and not an equality of subgroup images, that sends the defect to the identity.

Remark S7.3 (chain conditions suffice). Finiteness of the invariant can be weakened to a chain condition. A conjugation-equivariant monotone invariant whose values satisfy the ascending or descending chain condition takes the same value on Γ as on $t\Gamma t^{-1}$, because an order isomorphism of such a poset fixes every point it moves weakly upward. The fixed subspaces of Γ and $t\Gamma t^{-1}$ agree in every finite-dimensional representation over every field; the same holds for the invariants of the induced representations on $V^{\otimes p} \otimes (V^*)^{\otimes q}$, of which the commutant identity (4) is the case $\text{End}(V) \cong V \otimes V^*$; and the Zariski closures of $\sigma(\Gamma)$ and $\sigma(t\Gamma t^{-1})$ coincide, the descending chain condition for Zariski-closed subgroups of $\text{GL}(V)$ coming from the Hilbert basis theorem.

S8. THE CYCLIC COMPARISON

Theorem S8.1 (the cyclic comparison). *Define the presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_{BS} is finitely presented and $w_{\text{BS}} \neq 1$. There is an explicit surjective homomorphism from E_{BS} onto the subgroup of the Clifford semidirect product generated by the displayed images, and it maps w_{BS} to the nontrivial central sign. Nevertheless every finite-dimensional representation of E_{BS} over every field maps w_{BS} to the identity.

Thus being sent to the identity by every exact finite-dimensional representation does not by itself force triviality of the distinguished word.

Proof of Theorem S8.1. As with w , the relator $w_{\text{BS}}^2 = 1$ is redundant: tct^{-1} and $\gamma_0(tct^{-1})\gamma_0^{-1}$ are involutions, so centrality forces it by (5). Deleting it gives a six-relator presentation of the same group. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. The same commutant argument as in Theorem 3.3 therefore maps w_{BS} to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of $\text{BS}(1, 2)$ and its coset action. The base point is fixed by the cyclic base, while the two points represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these points and map c to the generator at the base point. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\text{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjection onto a Clifford quotient. $\square$

Sharpness of the Kazhdan hypothesis. The Clifford quotient in Theorem S8.1 is amenable, hence MF, and its quotient map sends w_{BS} to a nontrivial element. Composing with a faithful homomorphism from that quotient to $\mathcal{U}(\mathcal{Q})$ gives a homomorphism $\Theta: E_{\text{BS}} \rightarrow \mathcal{U}(\mathcal{Q})$ with $\Theta(w_{\text{BS}}) \neq 1$, so property (T) cannot be omitted from the central involution obstruction. In the residual language of Section S10, $w_{\text{BS}} \notin \text{Res}_{\text{MF}}(E_{\text{BS}})$. The affine group $\text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ is solvable. The Clifford group on its coset space is locally finite, so their semidirect product is amenable. That quotient is a subgroup of this amenable group and is therefore amenable, hence MF by [TWW]. Its distinguished word is the nontrivial Clifford sign by the calculation above.

The homomorphism to $\mathcal{U}(\mathcal{Q})$ just constructed cannot be assembled from exact finite-dimensional representations: it is an exact homomorphism, but any coordinate lifts of it are asymptotically multiplicative without being multiplicative. Indeed, neither E_{BS} nor its Clifford quotient is LEF, and the latter is not residually finite. In every finite group the doubling relation makes the image of $\langle\gamma_0^2\rangle$ equal to the image of $\langle\gamma_0\rangle$; the generator therefore commutes with the required “halved” element and the distinguished word maps to the identity, so neither group is residually finite. Any exact local embedding of a finite set containing the defining relators and the distinguished word satisfies the same equations, so an exact finite-dimensional representation would already send the distinguished word to the identity: neither group is LEF.

S9. THE SCALING FAMILY

For $m \geq 2$, let E_m be the same eight-generator presentation, replacing the three relations $tv_it^{-1} = v_i^2$ by $tv_it^{-1} = v_i^m$ and leaving every other relator unchanged; thus $E_2 = E$. Write w_m for the image of the same distinguished word w .

Corollary S9.1 (the scaling family). *For every natural number $m \geq 2$, the group E_m is finitely presented and w_m is a nontrivial central involution mapped to the identity by every homomorphism $E_m \rightarrow \mathcal{U}(\mathcal{Q})$. Consequently E_m is not MF, and neither $C_{\max}^*(E_m)$ nor $C_r^*(E_m)$ is an MF C^* -algebra.*

Proof. Replace the affine dilation $\text{diag}(2, 2, 2, 1)$ by $\text{diag}(m, m, m, 1)$ (both acting as in Lemma S2.2). It conjugates each integral translation v_i to v_i^m and fixes the three linear generators. Its inverse sends v_1 to translation by e_1/m , which is not integral; hence the two Clifford generators remain distinct and w_m maps to -1 . The Kazhdan base, the one-sided conjugation argument, and the central involution obstruction are independent of m . $\square$

S10. THE MF RESIDUAL

Definition S10.1. For a countable group G_1 , the *MF residual* is

$$\text{Res}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a homomorphism } G_1 \rightarrow \mathcal{U}(\mathcal{Q}) \},$$

a normal subgroup of G_1 . If G_1/N is MF then composing $G_1 \rightarrow G_1/N$ with an injective homomorphism $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ gives a homomorphism with kernel N , so $\text{Res}_{\text{MF}}(G_1) \leq N$: it is the smallest normal subgroup with MF quotient, which is the sense in which *residual* is used here and for the nilpotent and finite residuals. The intersection does not change if the dimension sequences are allowed to be arbitrary positive rather than strictly increasing (Proposition 2.3). This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

Lemma S10.2 (functoriality). *For every homomorphism $\varphi: G_1 \rightarrow G_2$ of countable groups,*

$$\varphi(\text{Res}_{\text{MF}}(G_1)) \subseteq \text{Res}_{\text{MF}}(G_2) :$$

the MF residual is functorial. In particular, if $\varphi: E \rightarrow G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \text{Res}_{\text{MF}}(G_1)$ and Θ is a homomorphism $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$, then $\Theta \circ \varphi$ is a homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$, so $\Theta(\varphi(x)) = 1$. For the second statement, $w \in \text{Res}_{\text{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \text{Res}_{\text{MF}}(G_2)$, and an injective homomorphism $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$ would map $\varphi(w)$ to 1, a contradiction. $\square$

Proposition S10.3 (the universal MF quotient). *Let G_1 be a countable group and $R = \text{Res}_{\text{MF}}(G_1)$.*

- (1) *R is the kernel of a single homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$.*

(2) G_1/R is MF.

(3) Every homomorphism from G_1 to an MF group factors through G_1/R .

In particular G_1 is MF if and only if $R = \{1\}$.

Proof. (3) If $\varphi: G_1 \rightarrow M$ with M MF and $j: M \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$ is an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, then $j \circ \varphi$ is a homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$, so $j(\varphi(R)) = 1$ and, by injectivity of j , $\varphi(R) = 1$.

(1) and (2). Work on $Q = G_1/R$. Every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ maps R to 1, and for each $g \notin R$ there is one with $\Theta(g) \neq 1$, which descends; so for every $\bar{g} \neq 1$ in Q there is a homomorphism $\Theta_{\bar{g}}: Q \rightarrow \mathcal{U}(\mathcal{Q})$ with $\Theta_{\bar{g}}(\bar{g}) \neq 1$. Enumerate the nonidentity elements and these representations, pass to coordinate unitary lifts, and diagonalize: at stage k , select in each of the first k of them a coordinate at which multiplicativity holds within $1/k$ on the first k group elements and each of the first k designated elements stays separated by its fixed positive constant — possible because each $\Theta_{\bar{g}}$ separates its own element along infinitely many coordinates — and take the block sum. Block operator norms are maxima, so the resulting sequence is an operator-norm asymptotic representation separating every nonidentity element from the identity; its class, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful homomorphism $Q \rightarrow \mathcal{U}(\mathcal{Q})$. This proves (2), and composing with $G_1 \twoheadrightarrow Q$ proves (1). Finally, if G_1 is MF then an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ gives $R = \{1\}$; if $R = \{1\}$ then (2) says G_1 is MF. $\square$

Corollary S10.4 (exact residual from a candidate quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Res}_{\text{MF}}(G_1)$. Every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ factors uniquely through G_1/N . If in addition G_1/N is MF, then*

$$\text{Res}_{\text{MF}}(G_1) = N.$$

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every such Θ . For the reverse containment in the displayed equality, compose the quotient map $G_1 \rightarrow G_1/N$ with an injective homomorphism $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$: its kernel is exactly N , while the MF residual lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$. $\square$

Corollary S10.5 (residual reduction to the quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Res}_{\text{MF}}(G_1)$ and let $q: G_1 \rightarrow G_1/N$ be the quotient map. Then*

$$\text{Res}_{\text{MF}}(G_1) = q^{-1}(\text{Res}_{\text{MF}}(G_1/N)).$$

Proof. For the forward containment, Lemma S10.2 gives $q(\text{Res}_{\text{MF}}(G_1)) \subseteq \text{Res}_{\text{MF}}(G_1/N)$. Conversely, suppose $q(x) \in \text{Res}_{\text{MF}}(G_1/N)$ and let Θ be a homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$. Every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ maps N to 1 by hypothesis, so Θ factors as $\bar{\Theta} \circ q$ with $\bar{\Theta}$ a homomorphism $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$; hence $\Theta(x) = \bar{\Theta}(q(x)) = 1$. $\square$

By Corollary S10.5, the passage from N to the full preimage of the quotient residual is a closure operation on the normal subgroups of a fixed group.

Proposition S10.6 (the MF closure of relations). *For a countable group G_1 and $N \trianglelefteq G_1$ with quotient map $q: G_1 \rightarrow G_1/N$, put*

$$\text{cl}_{\text{MF}}(N) = q^{-1}(\text{Res}_{\text{MF}}(G_1/N)).$$

An element belongs to $\text{cl}_{\text{MF}}(N)$ exactly when every homomorphism $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ that maps N to the identity maps it to the identity. The operation $N \mapsto \text{cl}_{\text{MF}}(N)$ is extensive, monotone, and idempotent on normal subgroups; $G_1/\text{cl}_{\text{MF}}(N)$ is MF; and $\text{cl}_{\text{MF}}(N) = N$ exactly when G_1/N is MF.

Proof. Homomorphisms $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ that map N to 1 correspond to homomorphisms $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$, which gives the membership characterization; extensivity, monotonicity, and idempotence are read off from it. The quotient clause is Proposition S10.3(2) transported along the third isomorphism theorem, and the fixed-point clause is the final equivalence of Proposition S10.3. $\square$

For a presentation $G = \langle X \mid R \rangle = F/N$, the closure of N consists of the relations satisfied by every solution of R in every MF group, and the group is MF exactly when N is closed. The distinguished word of Theorem A lies in $\text{cl}_{\text{MF}}(N) \setminus N$ for the forty-one relators (Proposition S13.2).

Proposition S10.7 (iterating closure and join). *Let $\mathcal{S}$ be the smallest family of subgroups of G_1 that contains the trivial subgroup and is closed under the following two operations: from a normal $N \in \mathcal{S}$, adjoin every subgroup of $\text{cl}_{\text{MF}}(N)$; and from any subfamily of $\mathcal{S}$, adjoin its join, the subgroup generated by the union. Then every member of $\mathcal{S}$ is contained in $\text{Res}_{\text{MF}}(G_1)$; a normal member with MF quotient equals $\text{Res}_{\text{MF}}(G_1)$; and if $G_1 \in \mathcal{S}$, every homomorphism from G_1 to a countable MF group is trivial.*

Proof. Induct over the two operations. If a normal $N \in \mathcal{S}$ satisfies $N \leq \text{Res}_{\text{MF}}(G_1)$, then Corollary S10.5 gives $\text{cl}_{\text{MF}}(N) = \text{Res}_{\text{MF}}(G_1)$, so every subgroup chosen in the first operation is again inside the residual — normal or not. For the second operation, the residual is itself a subgroup, hence an upper bound in the subgroup lattice for any family contained in it; so the join $\bigvee_i H_i = \langle \bigcup_i H_i \rangle$ lies in the residual too. The remaining assertions are Corollary S10.4 and Proposition S10.3; the latter needs N normal so that G_1/N is a group. $\square$

By Theorems S6.3, S6.6 and S11.2, a subgroup lies inside the residual of every countable group with the relevant data, hence inside the residual of every quotient reached by the iteration. A hypothesis may hold at a later stage without holding at an earlier one: modulo the earlier stage an element may acquire finite order, an orbit may begin to commute, or a subgroup may become normal.

Corollary S10.8. *Let A be a unital C^* -algebra admitting an injective, possibly nonunital, $*$ -homomorphism into a matrix quotient. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.*

Proof. Let $j: A \rightarrow \mathcal{Q}$ be the possibly nonunital inclusion into the ambient matrix quotient and let $\phi: E \rightarrow \mathcal{U}(A)$. Complement correction gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

This is an injective homomorphism on $\mathcal{U}(A)$, so $\hat{j} \circ \phi$ is a homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$. By Theorem A, $(\hat{j} \circ \phi)(w) = 1$; injectivity of $\hat{j}$ gives $\phi(w) = 1$. Proposition S2.3 has $w \neq 1$, so ϕ cannot be injective. $\square$

S10.1. Permanence and its failure.

Lemma S10.9 (permanence). *Let all groups be countable.*

- (1) *Every subgroup of an MF group is MF.*
- (2) *Every residually finite group is MF.*
- (3) *Every locally finite group is MF.*

Proof. (1) Restrict an injective homomorphism to $\mathcal{U}(\mathcal{Q})$. (2) For a finite test set F , a finite quotient injective on F , composed with its left regular permutation action, is an exact finite-dimensional representation; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-global direction of Proposition 2.3 concludes. (3) A finite test set lies in a finite subgroup K . The left regular representation $\lambda_K: K \rightarrow \mathcal{U}(\mathbb{C}^K)$ is a finite-dimensional unitary representation of K by permutation matrices, and the local model that sends each test element g to $\lambda_K(g)$ is exactly multiplicative on the test set, since a product of test elements again lies in K . It separates as in (2). $\square$

Corollary S10.10 (quotients). *MF groups are not closed under quotients: the free group F_8 is MF, and the explicit group E is its quotient but is not MF.*

Proof. Map the eight free generators onto the eight displayed generators of E . The resulting homomorphism is surjective because those generators define the displayed presentation. The group F_8 is residually finite, hence MF by Lemma S10.9(2), while E is not MF by Theorem A. $\square$

S10.2. Matricial stability and the residual. Following Dadarlat [Dad], a countable group G is *matricially stable* if for every operator-norm asymptotic representation (V_n) of G there are genuine finite-dimensional unitary representations π_n , in the same matrix dimensions, with $\|V_n(g) - \pi_n(g)\| \rightarrow 0$ for every $g \in G$. Write $\text{Res}_{\text{fd}}(G)$ for the intersection of the kernels of all finite-dimensional unitary representations of G .

Proposition S10.11 (stability identifies the residuals). *For every countable group, $\text{Res}_{\text{MF}}(G) \leq \text{Res}_{\text{fd}}(G)$. If G is matricially stable, then $\text{Res}_{\text{MF}}(G) = \text{Res}_{\text{fd}}(G)$; in particular a stable group with $\text{Res}_{\text{fd}}(G) \neq 1$ is not MF.*

Proof. The first inclusion holds for every countable group. A finite-dimensional unitary representation π on $\mathbb{C}^d$ amplifies to $\pi^{\oplus n}$ on $\mathbb{C}^{nd}$; the dimensions nd are strictly increasing, so $(\pi^{\oplus n})_n$ is a homomorphism to $\mathcal{U}(\mathcal{Q})$

in the sense of (3). Since $\|\pi^{\oplus n}(x) - 1\| = \|\pi(x) - 1\|$ for every n , that sequence is null exactly when $\pi(x) = 1$. Hence an element of $\text{Res}_{\text{MF}}(G)$ is sent to the identity by every finite-dimensional representation.

For the converse inclusion, assume G matricially stable, let $x \in \text{Res}_{\text{fd}}(G)$, and suppose toward a contradiction that a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ has $\Theta(x) \neq 1$. Choose unitary coordinate lifts $U_n(g)$ of $\Theta(g)$. Since $\Theta(x) - 1 \neq 0$ in the norm quotient, after passing to a subsequence there is $\delta > 0$ with $\|U_n(x) - 1\| \geq \delta$ for every n . The lifts form an operator-norm asymptotic representation, so stability gives genuine representations π_n on the same blocks with $\|U_n(g) - \pi_n(g)\| \rightarrow 0$ for each fixed g . But $x \in \text{Res}_{\text{fd}}(G)$ gives $\pi_n(x) = 1$ for every n , so $\|U_n(x) - 1\| = \|U_n(x) - \pi_n(x)\| \rightarrow 0$, against the uniform lower bound. $\square$

For a stable group the two residuals therefore agree. With Theorem 3.3: every matricially stable group with $\text{Res}_{\text{fd}}(G) \neq 1$ is not MF.

S11. ORBIT COLLAPSE

Under a commuting-orbit hypothesis in place of centrality, every homomorphism to $\mathcal{U}(\mathcal{Q})$ identifies an entire conjugation orbit of an involution; a whole family of commutators therefore lies in the residual, and by Corollary S10.5 computing the residual reduces to the quotient by that family.

Definition S11.1 (involutive orbit witness). Let H be a group, $L \leq H$ a subgroup, and $s \in H$. An element $k \in H$ is an *involutive orbit witness* for (L, s) if

- (W1) $k^2 = 1$;
- (W2) $s\gamma s^{-1}$ commutes with k for every $\gamma \in L$;
- (W3) the conjugates $\gamma_1 k \gamma_1^{-1}$ and $\gamma_2 k \gamma_2^{-1}$ commute for all $\gamma_1, \gamma_2 \in L$.

The pair (L, s) determines the normal subgroup

$$D_{\text{coll}}(L, s) = \langle\langle [\gamma, k] : \gamma \in L, k \text{ a witness for } (L, s) \rangle\rangle_H.$$

In a permutational wreath product $K^{(X)} \rtimes G$ with $L \leq G$ conjugated into itself by s and $x_0 \in X$ a point fixed by sLs^{-1} , every involution of the copy of K at x_0 is a witness: copies at distinct points commute, and the conjugate copy fixes the point. Every homomorphism to $\mathcal{U}(\mathcal{Q})$ then identifies those copies along the whole L -orbit of x_0 (Theorem S11.2).

Theorem S11.2 (involutive collapse). *Let H be countable, let $L \leq H$ be a subgroup with property (T), and let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$. Then*

$$D_{\text{coll}}(L, s) \leq \text{Res}_{\text{MF}}(H).$$

Proof. Fix a witness k and suppose, toward a contradiction, that some homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ has $\Theta([\gamma_0, k]) \neq 1$ for some $\gamma_0 \in L$. Coordinate unitary lifts (Lemma 2.2) give an operator-norm asymptotic representation

$(U_{g,n})$ on blocks $M_{d_n}(\mathbb{C})$, and after passing to a subsequence there is $\eta > 0$ with $\|U_{[\gamma_0,k],n} - 1\| \geq \eta$ for every n .

Step 1: approximation by commuting self-adjoint involutions. Write $v_\gamma = \gamma k \gamma^{-1}$ and let $O = \{v_\gamma : \gamma \in L\}$ be the conjugation orbit of k ; by (W1), (W3), and asymptotic multiplicativity the models of the elements of O asymptotically commute and square to 1. The spectral sign of the Hermitian part of a unitary u is an exact self-adjoint involution at operator distance at most $\|u^2 - 1\|$ from u . Enumerate O as $v^{(1)}, v^{(2)}, \dots$ and correct in that order, carrying the induction hypothesis that after stage m the operators $V_n(v^{(1)}), \dots, V_n(v^{(m)})$ are exact self-adjoint involutions commuting pairwise, *and that no later stage alters them*. At stage m the model $U_{v^{(m)},n}$ fails to commute with $V_n(v^{(j)})$, $j < m$, by at most $\|[U_{v^{(m)},n}, U_{v^{(j)},n}] - \| + 2\|V_n(v^{(j)}) - U_{v^{(j)},n}\|$, the first term an asymptotic-commutation error of (W3) and the second the error already accumulated, so the total is a sum of m quantities each tending to 0. The joint commutant of $V_n(v^{(1)}), \dots, V_n(v^{(m-1)})$ is a unital $*$ -subalgebra — a finite intersection of joint spectral corners of exact commuting involutions — so compressing $U_{v^{(m)},n}$ into it moves it by at most that amount, and taking the spectral sign of the Hermitian part *inside that subalgebra* yields an exact self-adjoint involution commuting exactly with all earlier ones while leaving them untouched. The resulting perturbation is bounded by the sum of the preceding m commutation and correction errors, hence tends to 0 for fixed m ; each m is fixed before n grows, so the limits displayed next are taken at fixed m . The induction yields exact pairwise-commuting self-adjoint involutions $V_n(v)$, $v \in O$, with

$$\|V_n(v) - U_{v,n}\| \longrightarrow 0, \quad \left\| U_{g,n} V_n(v) U_{g,n}^* - V_n(g v g^{-1}) \right\| \longrightarrow 0$$

for each fixed $v \in O$ and $g \in L$. Write $V_n(\gamma)$ for $V_n(v_\gamma)$; it depends on γ only through v_γ .

Step 2: a rank pseudometric on the orbit. For commuting exact involutions e, f the self-adjoint displacement $T = e - f$ satisfies $T^3 = 4T$, so its eigenvalues lie in $\{0, \pm 2\}$, the operator $T^2/4$ is the orthogonal projection onto $\text{ran } T$, and

$$(11) \quad \text{Tr } |e - f|^2 = 4 \text{rank}(e - f).$$

Hence $r_n(\gamma, \mu) = \text{rank}(V_n(\gamma) - V_n(\mu))$ is an integer-valued pseudometric. It is eventually translation invariant, $r_n(g\gamma, g\mu) = r_n(\gamma, \mu)$ for all large n and each fixed triple: the comparison is made between the projections $T^2/4$, whose ranks agree once they lie at operator distance below 1, and the displacements have operator norm at most 2, so a covariance error below $1/8$ moves the associated projections by at most $1/4$.

Step 3: normalization at the generators. Property (T) makes L finitely generated; fix a Kazhdan pair (S, κ) with S finite symmetric and put $k_n = \sum_{a \in S} r_n(1, a)$. For each fixed γ , translation invariance and the triangle

inequality along one fixed word for γ give $r_n(1, \gamma) \leq |\gamma|_S k_n$ for all large n , where $|\cdot|_S$ is the word length. Suppose $k_n = 0$ along a subsequence. Only γ_0 is needed there, and the estimate above is available for it: $r_n(1, \gamma_0) \leq |\gamma_0|_S k_n = 0$ at every sufficiently late stage of that subsequence, so $V_n(\gamma_0) = V_n(1)$ and $\|U_{[\gamma_0, k], n} - 1\| \rightarrow 0$ along it, against the separation. Hence $k_n \geq 1$ for all large n . (The bound $r_n(1, \gamma) \leq |\gamma|_S k_n$ holds for each fixed γ past a stage depending on γ , so it does not collapse the whole orbit at one stage when L is infinite; the argument does not ask it to.)

Step 4: an approximate 1-cocycle. This step and the next work with the *unnormalized* Hilbert–Schmidt, or Frobenius, norm $\|x\|_F = \text{Tr}(x^*x)^{1/2}$, so that $\|x\|_F^2 = d_n \|x\|_2^2$ on $M_{d_n}(\mathbb{C})$; the normalization is by the rank k_n below rather than by d_n . Set $b_n(\gamma) = k_n^{-1/2}(V_n(\gamma) - V_n(1))$. By (11) and Steps 2 and 3, $\|b_n(\gamma)\|_F^2 = 4r_n(1, \gamma)/k_n \leq 4|\gamma|_S$ at every large stage, and

$$(12) \quad \sum_{a \in S} \|b_n(a)\|_F^2 = 4$$

whenever $k_n \geq 1$. Conjugation by $U_{a,n}$ is a Frobenius isometry. Fix $a, \gamma \in L$, put $\mu = a^{-1}\gamma$, and let $k_n^{-1/2}D_n$ be the defect in the cocycle identity, so that

$$D_n = (V_n(\gamma) - V_n(1)) - U_{a,n}(V_n(\mu) - V_n(1))U_{a,n}^* - (V_n(a) - V_n(1)).$$

As written, D_n is a sum of three displacements of the commuting family of Step 1, one of them conjugated; rank is subadditive and invariant under unitary conjugation, so Step 2 with the bound of Step 3 gives

$$\text{rank } D_n \leq r_n(\gamma, 1) + r_n(\mu, 1) + r_n(a, 1) \leq (|\gamma|_S + |\mu|_S + |a|_S)k_n$$

eventually. Regrouped as

$$D_n = (V_n(\gamma) - U_{a,n}V_n(\mu)U_{a,n}^*) + (U_{a,n}V_n(1)U_{a,n}^* - V_n(a)),$$

the same D_n is a sum of two covariance errors of Step 1, so $\|D_n\| \rightarrow 0$. Both bounds are needed, because operator-norm control alone does not bound rank: two exact involutions at small operator distance can differ in full rank. Combining them in $\text{Tr } |D|^2 \leq \text{rank}(D) \cdot \|D\|^2$ gives $\text{Tr } |D_n|^2/k_n \leq (|\gamma|_S + |\mu|_S + |a|_S) \|D_n\|^2 \rightarrow 0$: the cocycle defect is $o(k_n)$ in squared Frobenius norm.

Step 5: the limiting cocycle is a coboundary. Fix a free ultrafilter ω on $\mathbb{N}$, give $M_{d_n}(\mathbb{C})$ the inner product $\langle x, y \rangle = \text{Tr}(y^*x)$, and let K_ω be the Hilbert-space ultraproduct of these spaces along ω . Conjugation is unitary for that inner product, and the dimension-free estimate $\|\text{Ad } u_1 - \text{Ad } u_2\| \leq 2\|u_1 - u_2\|$ makes $g \mapsto \text{Ad } U_{g,n}$ asymptotically multiplicative in the operator norm of $B(M_{d_n}(\mathbb{C}))$, so

$$\pi(g) = [\text{Ad } U_{g,n}]_\omega$$

is a unitary representation of H on K_ω . The bound $\|b_n(\gamma)\|_F^2 \leq 4|\gamma|_S$ of Step 4 makes $b(\gamma) = [b_n(\gamma)]_\omega$ a well-defined vector of K_ω for every $\gamma \in L$,

and the vanishing of the normalized cocycle defect makes

$$b(a\mu) = b(a) + \pi(a)b(\mu) \quad (a, \mu \in L),$$

so $b: L \rightarrow K_\omega$ is a 1-cocycle for $\pi|_L$. Since $\lim_\omega$ is linear, (12) passes to the limit as the identity

$$(13) \quad \sum_{a \in S} \|b(a)\|^2 = 4.$$

By the Delorme–Guichardet theorem [BHV, Theorem 2.12.4], property (T) of L makes every 1-cocycle of every unitary representation a coboundary, so there is $y \in K_\omega$ with

$$b(\gamma) = y - \pi(\gamma)y \quad (\gamma \in L).$$

Step 6: transport by the conjugator s . By (W2) the element sas^{-1} commutes with k for every $a \in L$, so $v_{sas^{-1}} = k = v_1$; the assignment $\gamma \mapsto v_\gamma$ of Step 1 then gives $V_n(sas^{-1}) = V_n(1)$ at every stage, and hence $b(sas^{-1}) = 0$. Therefore $y = \pi(sas^{-1})y$ for every $a \in L$: the vector y is fixed by $\pi(sLs^{-1})$.

The proof of Theorem 3.4 gives the reverse of the inclusion supplied by $sLs^{-1} \leq L$, namely

$$\text{Fix } \pi(sLs^{-1}) \subseteq \text{Fix } \pi(L) :$$

there P is the projection of K_ω onto $\text{Fix } \pi(L)$, the range of $Q = \pi(s)P\pi(s)^*$ is $\text{Fix } \pi(sLs^{-1})$, $P \leq Q$, and $Q = P$ by Lemma 3.1 and finiteness of B_ω . That chain is independent of (x_n) , and multiplying each coordinate inner product by a positive scalar — here passing from tr_{d_n} to Tr — changes neither the operator norms of the adjoint actions nor the comparison of projections, so it applies unchanged.

Hence $y \in \text{Fix } \pi(L)$, so $b(a) = y - \pi(a)y = 0$ for every $a \in L$, contradicting (13). Therefore $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$ and every witness k , and normality of these kernels extends the conclusion to all of $D_{\text{coll}}(L, s)$. $\square$

Remark S11.3 (the quantitative form). Steps 5 and 6 can be carried out at a single coordinate, with every tolerance fixed in advance, except for the constant C below, which is read off the limiting cocycle. By property (T), that *limiting* cocycle satisfies a bound uniform in γ ,

$$(14) \quad \|b(\gamma)\| \leq C \quad (\gamma \in L),$$

with C depending on the Kazhdan pair alone. Step 4 gives only the γ -dependent bound $2|\gamma|_S^{1/2}$; (14) strengthens it to a bound uniform in γ .

The finite-stage reading of (14) fails, and the argument below does not use it. For each fixed γ the finite-stage norms $\|b_n(\gamma)\|_F$ approach $\|b(\gamma)\|$ along ω , so each γ has its own ω -large set of good stages; there is no single stage at which the bound holds simultaneously for all $\gamma \in L$, since L is infinite and an infinite intersection of ω -large sets need not be ω -large. The bound is used at finitely many places only — on the generators $a \in S$ and on the primitive — and there the intersection is finite. Property (T) supplies that uniformity, in the quantitative form of the Delorme–Guichardet argument

[BHV, Section 2.12]: for a Kazhdan pair (S, κ) , every 1-cocycle of every unitary representation of L is bounded by its generator values, the fixed vector produced from the Kazhdan projection lying within a multiple of $\max_{a \in S} \|b(a)\|$ of the origin and $b(\gamma) = y - \pi(\gamma)y$ being at most twice that. Applied to the limiting cocycle of Step 5, where $\sum_{a \in S} \|b(a)\|^2 = 4$ by (12), this gives

$$(15) \quad \|b(\gamma)\|^2 \leq \frac{160 \log 2}{\kappa^2} \leq \frac{111}{\kappa^2} \quad (\gamma \in L),$$

so $C = \sqrt{111}/\kappa$ serves. The Kazhdan constant is the only datum that enters: neither $|S|$ nor the group appears, which is more than uniformity in γ and more than (14) asks for. The constant C is the only quantity below taken from the limit; the transport and the contradiction take place at a single stage. Put

$$\rho = \frac{1}{\sqrt{|S|} + 1}, \quad q = \frac{\rho}{24(C + 1)}, \quad \tau \leq \min\left\{\frac{\rho}{2}, \frac{(1 - \theta)\rho^2}{256(C + 1)}\right\},$$

with

$$1 - \frac{\kappa^2}{4|S|} < \theta < 1$$

the spectral threshold, and fix a single stage n , past the finitely many thresholds of Steps 1–4, at which the covariance errors of Step 1 are below τ . Both inequalities on θ are used. The upper one keeps the corner nonempty. The lower one holds because the Kazhdan gap of Step 4 bounds the Hermitian average by $1 - \kappa^2/(4|S|)$ off the fixed subspace, and it yields (16).

The operator to cut is not quite the average. At a finite stage $|S|^{-1} \sum_{a \in S} \text{Ad } U_{a,n}$ need not be self-adjoint even for symmetric S , because $U_{a^{-1},n}$ is only *approximately* $U_{a,n}^*$; an ordered spectral cut “above θ ” is therefore undefined for it as it stands. Let P_n be instead the spectral projection above θ of the Hermitian part

$$\frac{1}{2|S|} \sum_{a \in S} (\text{Ad } U_{a,n} + (\text{Ad } U_{a,n})^*)$$

— the *Kazhdan corner* — and let $\widehat{b}_n$ be the compression into it of the primitive supplied below.

$$(16) \quad \|(\text{Ad } U_{a,n} - 1)P_n\| \leq q \quad (a \in S),$$

$$(17) \quad \|(1 - P_n)Q_n\| \leq q, \quad Q_n = \text{Ad } U_{s,n} P_n \text{Ad } U_{s,n}^*.$$

The first says the corner is almost fixed, by the Kazhdan gap. The second is the finite-stage form of $P = \pi(s)P\pi(s)^*$. First the easy containment $\|(1 - Q_n)P_n\| \leq \varepsilon$: for $a \in S$ the one-sided relation gives $sas^{-1} = \delta$ for a single $\delta \in L$, and

$$(1 - \text{Ad } U_{a,n}) \text{Ad } U_{s,n}^* P_n = \text{Ad } U_{s,n}^* (1 - \text{Ad } U_{\delta,n}) P_n,$$

whose norm the Kazhdan gap makes small. Each δ is a word in S , and the gap controls the displacement it induces one letter at a time, so this is

one more threshold per generator — $|S|$ of them, all fixed before the stage is chosen. So the corner lies almost inside its image under s . Then the equal-rank reversal estimate of Section S4 turns that containment around: Q_n is a unitary conjugate of P_n and so has the same rank, and for two projections of equal finite rank a containment within $\varepsilon < 1$ in one direction gives one within $\varepsilon/\sqrt{1-\varepsilon^2}$ in the other. The estimate uses only finiteness: the leakage bound makes the compression $\text{ran } P_n \rightarrow \text{ran } Q_n$ injective, and an injective endomorphism of a finite-dimensional space is surjective. In infinite dimensions this last step fails, a proper isometry being injective and not surjective. The one-sided inclusion $sLs^{-1} \leq L$ enters as the only group-theoretic input.

The compressed object is a primitive for the displacement map b_n . The Delorme–Guichardet bound (14) on the limiting cocycle produces, at each tolerance, a stage n_0 and a matrix w_{n_0} with $\|w_{n_0}\|_F \leq C + 1$ satisfying

$$\|b_{n_0}(a) - (w_{n_0} - U_{a,n_0}w_{n_0}U_{a,n_0}^*)\|_F \leq \tau \quad (a \in S),$$

an approximate coboundary at one stage, for the finitely many generators only. Write $\hat{b}_n = P_n w_n P_n$ for its compression into the Kazhdan corner. The bound on w_{n_0} is the last use of the limiting cocycle; every estimate below concerns the single stage n_0 .

Compressing into the corner loses at most $c_{\text{cap}} = (C + 1)2\tau/(1 - \theta)$ in squared Frobenius norm, and one stage of transport bounds the displacement of $\hat{b}_n$ by each generator:

$$\text{Tr}|\hat{b}_n - U_{a,n}\hat{b}_nU_{a,n}^*|^2 \leq 18q^2C^2 + 16c_{\text{cap}} \leq \left(\frac{\rho}{2}\right)^2,$$

the second inequality being the arithmetic of the choice of q and τ . The primitive enters here, and it is the coboundary relation rather than proximity that closes the argument: $b_n(a)$ does not approximate $\hat{b}_n$ — the left side moves with a and the right side does not — but approximates its displacement,

$$\|b_n(a) - (\hat{b}_n - U_{a,n}\hat{b}_nU_{a,n}^*)\|_F \leq \tau \leq \frac{\rho}{2} \quad (a \in S).$$

Combined with the displayed transport bound this gives $\|b_n(a)\|_F \leq \rho$ for every $a \in S$ at once, while (12) reads $\sum_{a \in S} \|b_n(a)\|_F^2 = 4$ and $\sum_{a \in S} \rho^2 = |S|/(\sqrt{|S|} + 1)^2 < 1$: the single stage n already fails.

The commutation hypothesis (W3) is essential: it gives the joint spectral calculus behind the rank identity of Step 2. Anticommuting Clifford coordinates have no joint spectral labels; they are instead collapsed by the central sign of Theorem A. In $\mathcal{Q}$, Theorem S11.4 below removes the restriction to involutions, and Theorem S11.5 extends the collapse to witnesses of every finite order through their discrete Fourier idempotents.

The witness of Theorem S11.2 is a group element. The same collapse holds for arbitrary projections of $\mathcal{Q}$.

Theorem S11.4 (projection collapse). *Let H be countable, let $L \leq H$ have property (T), let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let*

$\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ be a homomorphism. Let $p \in \mathcal{Q}$ be a projection such that p commutes with $\Theta(s\gamma s^{-1})$ for every $\gamma \in L$ and the projections $\Theta(\gamma)p\Theta(\gamma)^*$, $\gamma \in L$, commute pairwise. Then p commutes with $\Theta(L)$.

Proof. The symmetry $\sigma = 1 - 2p$ is a self-adjoint unitary of $\mathcal{Q}$. The subgroup $H' \leq \mathcal{U}(\mathcal{Q})$ generated by $\Theta(H)$ and σ is countable. Inside H' , the image $\Theta(L)$ has property (T), $\Theta(s)$ conjugates it into itself, σ is an involution centralizing the conjugate copy, and the $\Theta(L)$ -conjugates of σ commute because the conjugated projections do. So σ is an involutive orbit witness in H' , so by Theorem S11.2 every commutator $[\Theta(\gamma), \sigma]$, $\gamma \in L$, lies in $\text{Res}_{\text{MF}}(H')$. The inclusion $H' \rightarrow \mathcal{U}(\mathcal{Q})$ is itself a homomorphism $H' \rightarrow \mathcal{U}(\mathcal{Q})$, so it maps each of these commutators to the identity: $[\Theta(\gamma), \sigma] = 1$ in $\mathcal{Q}$, and $\sigma = 1 - 2p$ gives $[\Theta(\gamma), p]_- = 0$. $\square$

Theorem S11.4 applies to an arbitrary projection of $\mathcal{Q}$, not only to one arising from a group element. So every projection $p \in \Theta(sLs^{-1})' \cap \mathcal{Q}$ whose $\Theta(L)$ -conjugates commute pairwise already lies in $\Theta(L)' \cap \mathcal{Q}$.

If k has finite order m , applying Theorem S11.4 to the Fourier spectral projections of $\Theta(k)$ removes the assumption that k is an involution.

Theorem S11.5 (torsion collapse). *Let H be countable, let $L \leq H$ have property (T), let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $k \in H$ have finite order. If the conjugate copy sLs^{-1} centralizes k and the conjugates $\gamma k \gamma^{-1}$, $\gamma \in L$, commute pairwise, then every homomorphism $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$; the normal closure of all such commutators, over all finite-order witnesses for (L, s) , lies in $\text{Res}_{\text{MF}}(H)$, with equality whenever the corresponding quotient of H is MF.*

Proof. Let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$, m the order of k , and $u = \Theta(k)$, a unitary with $u^m = 1$. Its discrete Fourier idempotents $p_j = m^{-1} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t / m} u^t$, for $j \in \mathbf{Z}/m$, are self-adjoint idempotents of $\mathcal{Q}$. Each p_j is a polynomial in u , so whatever commutes with u commutes with every p_j : the represented conjugate copy centralizes each p_j , and the $\Theta(L)$ -conjugates of each p_j commute pairwise because the conjugated witnesses do. Theorem S11.4 therefore makes every p_j commute with $\Theta(L)$, and Fourier reconstruction expresses $u = \sum_j e^{2\pi i j / m} p_j$, so u commutes with $\Theta(L)$ and $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. $\square$

In the quotient of the Kazhdan–Clifford group of Theorem A by its central involution the Clifford generators commute, and the moved base-point generator is itself an involutive witness.

Theorem S11.6 (the commuting-generator quotient). *The quotient $W/\langle \zeta \rangle$ of the Kazhdan–Clifford group by its central involution is sofic and is not MF. More precisely, the image of the commutator $[v_1, tct^{-1}]$, the inverse of the distinguished commutator u , in $W/\langle \zeta \rangle$ is nontrivial and lies in $\text{Res}_{\text{MF}}(W/\langle \zeta \rangle)$; that last conclusion holds with W replaced by the ambient affine–Clifford*

group of any injective endomorphism α of a countable property-(T) group and any base element outside the image of α .

Proof. Write $q: W \rightarrow W/\langle\zeta\rangle$ and $k = q(tct^{-1})$, the moved base-point generator. Since $q(\zeta) = 1$, any two Clifford generators commute in the quotient, so the $q(L)$ -conjugation orbit of k commutes pairwise, where $L = q(\bar{\Gamma})$ is the image of the base copy; k remains an involution, and the conjugate copy $q(t)Lq(t)^{-1}$ centralizes k because $q(t)Lq(t)^{-1}$ fixes the moved point. Property (T) passes to the image L , so by Theorem S11.2 every commutator $[q(\bar{v}_1), k]$ lies in the MF residual of the quotient. The commutator remains nontrivial in the sign quotient: in the signed finite-support model of $\text{Cl}(X)$, the commutator is the product of the two distinct basis elements at τo and $j(\bar{v}_1)\tau o$, whose configuration coordinate is nonzero, while every element of $\langle\zeta\rangle$ has configuration coordinate zero.

The Clifford kernel becomes abelian in the quotient: $\text{Cl}(X)/\langle\zeta\rangle$ is the elementary abelian group $\bigoplus_X \mathbb{Z}/2$ under the same Σ -action, so $W/\langle\zeta\rangle \cong (\bigoplus_X \mathbb{Z}/2) \rtimes \Sigma$. That kernel is again locally finite with finite level orbits, so the finite-by-sofic exhaustion in the proof of Theorem D proves soficity. $\square$

The same argument applies to the finitely presented group $E/\langle w \rangle$.

Proof of Theorem B. Write $q: E \rightarrow E/\langle w \rangle$, $\bar{d} = q(d)$, and $\bar{L} = q(\iota(\mathcal{B}))$. The element $\bar{d}$ is an involution because $c^2 = 1$, and $q(t)\bar{L}q(t)^{-1}$ centralizes $\bar{d}$ because $t\iota(\mathcal{B})t^{-1}$ commutes with c in E . Property (T) passes to $\bar{L}$, and $q(t)$ conjugates it into itself. Conditions (W1) and (W2) of Definition S11.1 therefore hold. It remains to verify (W3).

The linear generators x, y, z commute with d , and t carries each v_i to v_i^2 , so $\iota(v_i^2)$ commutes with d as well. Hence $\iota(g)d\iota(g)^{-1}$ depends only on the exponent vector of the translation part of g modulo 2, which leaves eight conjugates, indexed by $(\mathbb{Z}/2)^3$. By Theorem 5.1 the distinguished word is $w = [d, \iota(v_1)d\iota(v_1)^{-1}]$, the commutator attached to the class of v_1 . The six words x, x^2, z, yz, yx, y^2x in the linear generators carry that class onto the six remaining nonzero classes, and each commutes with d ; conjugating the displayed identity by the corresponding $\iota(r)$ exhibits every nonzero parity commutator as a conjugate of w , while the zero class contributes $[d, d] = 1$. Since w is central, all eight lie in $\langle w \rangle$, and one further conjugation reduces an arbitrary pair of conjugates to this case. Thus $\bar{d}$ is an involutive orbit witness for $(\bar{L}, q(t))$, and by Theorem S11.2 $[q(\iota(v_1)), \bar{d}] \in \text{Res}_{\text{MF}}(E/\langle w \rangle)$.

Nontriviality passes through the homomorphism $E \rightarrow W$ of Proposition S2.3: it carries $[\iota(v_1), d]$ to the product of the Clifford generators at the two distinct points τo and $j(\bar{v}_1)\tau o$, and carries $\langle w \rangle$ into $\langle\zeta\rangle$, whose configuration coordinate is zero. Hence $[\iota(v_1), d] \notin \langle w \rangle$, its image in $E/\langle w \rangle$ is nontrivial, and Proposition S10.3 makes $E/\langle w \rangle$ non-MF. Corollary S10.5 identifies $\text{Res}_{\text{MF}}(E)$ with the q -preimage of $\text{Res}_{\text{MF}}(E/\langle w \rangle)$, so u belongs to it. $\square$

Corollary S11.7 (collapse computes the residual). *In the setting of Theorem S11.2, write $q: H \rightarrow H/D_{\text{coll}}(L, s)$ for the quotient map. Then, unconditionally,*

$$\text{Res}_{\text{MF}}(H) = q^{-1}(\text{Res}_{\text{MF}}(H/D_{\text{coll}}(L, s))),$$

and if $H/D_{\text{coll}}(L, s)$ is MF, then $\text{Res}_{\text{MF}}(H) = D_{\text{coll}}(L, s)$.

Proof. Combine Theorem S11.2 with Corollaries S10.5 and S10.4. $\square$

In the second case $H/D_{\text{coll}}(L, s)$ is the universal MF quotient of H (Proposition S10.3): every homomorphism $H \rightarrow \mathcal{U}(\mathcal{Q})$ factors through it, and it is MF.

Theorem S11.8 (when the collapse subgroup is everything). *In the setting of Theorem S11.2:*

- (1) *Every homomorphism f from H to a countable MF group maps every element of $D_{\text{coll}}(L, s)$ to the identity.*
- (2) *If $D_{\text{coll}}(L, s) = H$, then $\text{Res}_{\text{MF}}(H) = H$; every homomorphism from H to a countable MF group is trivial; and if H is nontrivial, then H is not MF.*

Proof. (1) By Theorem S11.2 and Lemma S10.2,

$$f(D_{\text{coll}}(L, s)) \leq f(\text{Res}_{\text{MF}}(H)) \leq \text{Res}_{\text{MF}}(f(H)).$$

The subgroup $f(H)$ of the MF target is MF (Lemma S10.9(1)), so its residual is trivial by Proposition S10.3. (2) The residual statement is Theorem S11.2 with $D_{\text{coll}}(L, s) = H$; triviality of every homomorphism follows from (1); and a nontrivial group with full residual is not MF by Proposition S10.3. $\square$

For a group as in (2), every finite-dimensional unitary representation is trivial, every MF quotient is trivial, and every homomorphism to a countable residually finite group is trivial. Whether a small-cancellation construction produces a finitely presented group with $D_{\text{coll}}(L, s) = H$, or with a normal-Kazhdan defect equal to the whole group, is open.

S12. GROUP ALGEBRAS, SOFICITY, AND PERMANENCE

S12.1. The reduced group algebra of E . The stable finiteness in Theorem C comes from two standard facts.

Lemma S12.1. *Let G_1 be a countable discrete group.*

- (1) *The canonical tracial state τ on $C_r^*(G_1)$ is faithful.*
- (2) *Every unital C^* -algebra A with a faithful tracial state is stably finite: for every $k \geq 1$, every isometry in $M_k(A)$ is a unitary.*

Proof. (1) Realize $C_r^*(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g . Let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$. Each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C_r^*(G_1)$. The canonical trace is $\tau(x) = \langle x \delta_e, \delta_e \rangle$. If $\tau(x^* x) = 0$ then

$\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i -th diagonal entry of x^*x is $\sum_j x_{ji}^*x_{ji}$, so $\tau_k(x^*x) = 0$ implies $\tau(x_{ji}^*x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j . If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

S12.2. A sofic non-MF group.

Proof of Theorem D. Write $T = \varinjlim (\bar{\Gamma}, \bar{\alpha})$, so $\Sigma = T \rtimes \mathbb{Z}$, and let $B \leq \Sigma$ be the level-zero copy of $\bar{\Gamma}$. The image of $\bar{\alpha}$ has finite index (at most eight, by the parity classes of the translation lattice). Thus every level $\bar{\Gamma}_n \leq T$ is commensurable with B . The stable letter conjugates B onto $\bar{\alpha}(\bar{\Gamma})$, so all of Σ commensurates B . Orbit-stabilizer now shows that every $\bar{\Gamma}_n$ -orbit in $X = \Sigma/B$ is finite.

Consider a finite subset of $\text{Cl}(X) \rtimes T$. Its T -coordinates lie in one level $\bar{\Gamma}_n$. Each of its Clifford coordinates has finite $\bar{\Gamma}_n$ -orbit: this holds for the central sign and for each generator c_x , hence for every Clifford word. The finitely many resulting orbits generate a finite $\bar{\Gamma}_n$ -invariant subgroup $L \leq \text{Cl}(X)$, because $\text{Cl}(X)$ is locally finite. Our original finite set therefore lies in $L \rtimes \bar{\Gamma}_n$, which is sofic (finite-by-sofic). A group is sofic as soon as each of its finitely generated subgroups is, so $\text{Cl}(X) \rtimes T$ is sofic.

Reassociating the semidirect products gives

$$W = \text{Cl}(X) \rtimes (T \rtimes \mathbb{Z}) \cong (\text{Cl}(X) \rtimes T) \rtimes \mathbb{Z}.$$

Thus W is sofic, because a split extension of a sofic group by $\mathbb{Z}$ is sofic. Truncate the integer levels to $\mathbb{Z}/N$ and let the shift act cyclically. Induce the kernel's sofic approximations over this finite level set. The cocycle identity that makes the finite-quotient case exact now fails only on the wrap-around boundary. For a window shifted by k , that boundary occupies at most $2|k|/N$ of the levels, which vanishes as N grows. This is the $\mathbb{Z}$ -extension case of the Elek–Szabó amenable-extension theorem [ES06]. Here $\bar{\Gamma}$ is sofic: its six generators lie in $\text{GL}_4(\mathbb{Z})$, the resulting integral subgroup is residually finite through its congruence quotients, and it is isomorphic to $\bar{\Gamma}$.

The map $E \rightarrow W$ of Proposition S2.3 is surjective: its image contains Σ and the generator c_o at the base point, hence every Clifford generator by transitivity of the Σ -action on X . Since E is generated by v_1, x, y, z, t, c (by the Tietze calculation above), their six images generate W . The map sends $w \in \text{Res}_{\text{MF}}(E)$ to the nontrivial sign ζ ; functoriality (Lemma S10.2) makes $\zeta \in \text{Res}_{\text{MF}}(W)$. Hence W is not MF. $\square$

Failure of extension permanence. MF groups are not closed under extensions. More precisely, W fits into an exact sequence

$$1 \longrightarrow \mathrm{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel and quotient are MF, while W is not. The Clifford kernel is locally finite, LEF, sofic, and MF, while the group W is finitely generated and non-MF. Nearby approximation properties do have extension permanence of this shape: an extension of a residually finite group by a residually finite group is weakly sofic [GL], and weak soficity is detected by finite targets [GR].

Proof. By Lemma S10.9(3), the Clifford group is locally finite, hence MF. The group Σ identifies with the subgroup of $\mathrm{GL}_4(\mathbb{Q})$ generated by $\bar{\Gamma}$ and D : the levels are $D^{-n}\bar{\Gamma}D^n$, and the determinant gives the $\mathbb{Z}$ -coordinate. It is residually finite. At every odd modulus the determinant 8 of the dilation is invertible, so the congruence reduction of the integral affine model descends to conjugation by the dilation and extends over the direct limit and its shift. These reductions separate any nontrivial element; compare Mal'cev's theorem [Mal] for finitely generated linear groups. Thus Σ is MF by Lemma S10.9(2), whereas W is not MF by the preceding construction. $\square$

Extensions by $\mathbb{Z}$. Composing the two canonical projections gives a surjection $W \rightarrow \mathbb{Z}$ whose kernel is $\mathrm{Cl}(X) \rtimes T$, and this kernel is LEF, hence MF: every finite subset lies in $L \times \bar{\Gamma}_n$ for a finite invariant subgroup $L \leq \mathrm{Cl}(X)$ and one level, and that group is residually finite because the level acts on the finite group L through a finite quotient. Thus W is an extension of an LEF group by $\mathbb{Z}$ and is not MF, while soficity is preserved by every extension with amenable quotient, as in Elek–Szabó [ES06].

An exact stably finite non-MF algebra. The algebra $C_r^*(W)$ is separable, exact, stably finite, and not MF. Indeed, the locally finite Clifford kernel is amenable and hence exact, while the linear group $\Sigma \leq \mathrm{GL}_4(\mathbb{Q})$ is exact [GHW]. Exact groups are closed under extensions [KW2], so W is exact; for discrete groups this is equivalent to exactness of the reduced group algebra [KW1]. The canonical trace and stable finiteness follow as in Lemma S12.1. Finally, an MF embedding of $C_r^*(W)$ would restrict to a faithful homomorphism to $\mathcal{U}(\mathcal{Q})$ on the canonical group unitaries, contradicting that W is not MF.

A simple sofic envelope. There is a countable simple sofic group S_{simp} with

$$\mathrm{Res}_{\mathrm{MF}}(S_{\mathrm{simp}}) = S_{\mathrm{simp}}.$$

Equivalently, every homomorphism from S_{simp} to an MF group is trivial.

Proof. By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed W in such a group. Functoriality of the MF residual sends ζ to a nontrivial element of the MF residual of the simple group. That residual is normal, so it is the whole group. Every homomorphism to an MF group maps the MF residual to 1. $\square$

S12.3. Soficity of the literal group. Deleting the letter c and the families (C) and (W) from (9) leaves the presentation $\langle \mathcal{B}, t \mid tut^{-1} = \bar{\alpha}(u) \rangle$ of the ascending HNN extension of the base along the doubling endomorphism. By Remark S16.2 the base is $\bar{\Gamma}$, so

$$E/\langle\langle c \rangle\rangle \cong \Sigma$$

for the group $\Sigma = T \rtimes \mathbb{Z}$ of Section S2, with its site set $X = \Sigma/B$, base point o , shift τ and level-zero base $B = j(\bar{\Gamma})$. The levels

$$\Gamma_n = \tau^{-n} B \tau^n \quad (n \geq 0)$$

form an ascending chain of copies of $\bar{\Gamma}$ with union T . Put $K = \tau^{-1} B \tau$; then $B \leq K$ and $[K : B] = [\bar{\Gamma} : \bar{\alpha}(\bar{\Gamma})]$, at most eight. The cosets Σ/K are the *blocks*, and each block, as a set of sites, has $[K : B]$ elements.

Let $\mathcal{G}$ be the graph on X whose edges are the Σ -orbit of the pair $\{\tau o, j(\bar{v}_1)\tau o\}$, whose entries are distinct by Lemma S2.2. Two sites are adjacent in $\mathcal{G}$ exactly when they are distinct and lie in a common block. Adjacent sites share a block because $(\tau o)^{-1}(j(\bar{v}_1)\tau o) = \tau^{-1}j(\bar{v}_1)\tau \in K$. Conversely, the chart $u\bar{\alpha}(\bar{\Gamma}) \mapsto \tau^{-1}j(u)\tau o$ identifies the block of o with $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$ and the generating edge with the pair $\{\bar{\alpha}(\bar{\Gamma}), \bar{v}_1\bar{\alpha}(\bar{\Gamma})\}$; since $\bar{\alpha}(\bar{\Gamma})$ acts transitively on the nontrivial cosets of $\bar{\alpha}(\bar{\Gamma})$, every pair of distinct sites of a block is the image of the generating pair.

For a graph $\mathcal{G}$ on X , let $C(\mathcal{G})$ be the group presented by generators ζ and c_ξ ($\xi \in X$) and relations

$$\zeta^2 = c_\xi^2 = 1, \quad [\zeta, c_\xi] = 1, \quad [c_\xi, c_\eta] = \zeta \text{ for each edge } \{\xi, \eta\} \text{ of } \mathcal{G}.$$

For the complete graph on X this is $\text{Cl}(X)$ of Construction S2.1, and adjoining the anticommutation relations at the remaining pairs gives a surjection $C(\mathcal{G}) \rightarrow \text{Cl}(X)$.

Proposition S12.2 (normal form for E). *There is an isomorphism*

$$E \cong C(\mathcal{G}) \rtimes \Sigma$$

under which $\langle\langle c \rangle\rangle$ is the factor $C(\mathcal{G})$, the conjugate gcg^{-1} is the lamp $c_g B$, and the distinguished word w is the sign ζ . Each level Γ_n has finite orbits on X .

Proof. Send the six base letters and t to the corresponding generators of Σ , and c to c_o . Families (B) and (T) of (9) hold in Σ by construction. For family (C): $c_o^2 = 1$, and B fixes o , so each base generator fixes c_o . For family (W): the pair $\{\tau o, j(\bar{v}_1)\tau o\}$ is an edge, so the image of w is $[c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = \zeta$, which is central in $C(\mathcal{G})$ and fixed by Σ . This defines $E \rightarrow C(\mathcal{G}) \rtimes \Sigma$.

In the other direction, $\zeta \mapsto w$ and $c_g B \mapsto gcg^{-1}$ respects the defining relations of $C(\mathcal{G})$: the assignment is independent of the coset representative because the base centralizes c ; the images are involutions; w is central of order two; and along an edge $g\{\tau o, j(\bar{v}_1)\tau o\}$ the commutator of the two images is $gw g^{-1} = w$. Non-adjacent pairs carry no relation to verify. Combined with

the splitting $\Sigma \rightarrow E$ on the seven remaining letters, this gives $C(\mathcal{G}) \rtimes \Sigma \rightarrow E$. The two composites fix the eight letters of E and the generators ζ, c_ξ of $C(\mathcal{G})$, hence are the identity.

For the orbits, $\bar{\alpha}(\bar{\Gamma})$ has index at most eight in $\bar{\Gamma}$ — the parity classes of the translation lattice — so τ conjugates B onto a subgroup of finite index in itself. Every Γ_n is therefore commensurable with B , all of Σ commensurates B , and orbit–stabilizer makes each Γ_n -orbit in $X = \Sigma/B$ finite. $\square$

Composing the isomorphism with $C(\mathcal{G}) \rightarrow \text{Cl}(X)$ gives the surjection $E \rightarrow W$ of Proposition S2.3. The group $\text{Cl}(X)$ is locally finite; $C(\mathcal{G})$ is not, since two lamps at sites of different blocks generate an infinite dihedral group and three blocks give a nonabelian free subgroup. For a finite set S of sites write

$$M_S = \langle \zeta, c_\xi \ (\xi \in S) \rangle \leq C(\mathcal{G}).$$

Lemma S12.3 (windows are residually finite). *For every finite set S of sites the subgroup M_S is residually finite.*

Proof. Killing ζ sends $C(\mathcal{G})$ onto the free product of the elementary abelian 2-groups of the blocks, and M_S into the free product of the finitely many factors at blocks meeting S . A free product of finite groups is residually finite, and the products of lamps inside one block give a section over each factor, so the kernel of this map on M_S lies in $\langle \zeta \rangle$. The surjection $C(\mathcal{G}) \rightarrow \text{Cl}(X)$ sends ζ to the sign of $\text{Cl}(X)$, which is nontrivial, and $\text{Cl}(X)$ is locally finite, so the image of the finitely generated group M_S there is finite. An element of M_S killed by the first map lies in $\langle \zeta \rangle$, and the second map separates the two elements of that subgroup. $\square$

Proof of Theorem E. Let E_0 be the kernel of the homomorphism $E \rightarrow \mathbb{Z}$ given by the exponent sum in t . Under Proposition S12.2, $E_0 \cong C(\mathcal{G}) \rtimes T$.

Take a finite subset of E_0 . Its T -coordinates lie in one level Γ_n , the chain being ascending, and its $C(\mathcal{G})$ -coordinates lie in M_{S_0} for some finite set S_0 of sites, each generator of $C(\mathcal{G})$ being a single lamp or the sign. Let S be the union of the Γ_n -orbits of the sites in S_0 , a finite set by Proposition S12.2. Then M_S is Γ_n -invariant and contains M_{S_0} , so the finite subset lies in $M_S \rtimes \Gamma_n$.

That group is residually finite. Indeed M_S is residually finite by Lemma S12.3, the level $\Gamma_n \cong \bar{\Gamma}$ is residually finite through the congruence quotients of its matrix model (Lemma S2.2), and Γ_n acts on M_S through permutations of the finite set S , hence through a finite quotient F . The preimage of the identity of F is a subgroup of finite index isomorphic to $M_S \times \ker(\Gamma_n \rightarrow F)$, which is residually finite, and residual finiteness passes to a group containing a residually finite subgroup of finite index.

So E_0 is a directed union of subgroups embedding in residually finite groups, that is, LEF, and therefore sofic. Now E is a split extension of E_0 by $\mathbb{Z}$, split by t , so E is sofic by the Elek–Szabó amenable-extension theorem [ES06], and hyperlinear because sofic groups are. Failure of MF is

Theorem A. LEF groups are MF [CDE], so E is not LEF; a residually finite group is LEF, so E is not residually finite either. $\square$

S13. MARKED LIMITS, QUASI-IDENTITIES, AND UNDECIDABILITY

S13.1. Marked-limit closure. At every fixed number of marked generators the MF property is closed under limits of marked groups, and its failure has a finite certificate.

Theorem S13.1 (the MF locus is closed). *Fix $k \geq 1$ and consider the space of k -marked groups — quotients of the free group F_k , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators.*

- (1) *The marked groups with MF quotient form a closed set.*
- (2) *Equivalently, the non-MF marked groups form an open set.*
- (3) *Every non-MF k -marked group admits a finite-radius certificate: there is a radius R such that every k -marked group whose relations agree with it on the reduced words of length at most R is non-MF.*

Proof. A failure of MF for the quotient is a failure of normalized local approximation, and such a failure is witnessed on one finite subset of the quotient. Expressing the finitely many multiplications and inequalities of that subset by words in the marked generators produces finitely many relation constraints; every marked group agreeing on those constraints inherits the same failed local-model problem, so an entire basic clopen neighborhood fails MF. Enclosing the constraint words in a reduced-word ball gives the radius form. $\square$

S13.2. A finite universal Horn obstruction. Every MF group satisfies a single finite quasi-identity: a universal implication between finitely many equations, that is, a universal Horn sentence. Write R for the forty-one relators of (9), regarded as words in eight variables, and w for the distinguished word in those variables.

Proposition S13.2 (finite quasi-identity). *Every MF group G satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8: \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

while the canonical generating tuple of E satisfies every relator in R and has $w \neq 1$.

Proof. A tuple satisfying the relators induces a homomorphism $E \rightarrow G$ sending the distinguished word of E to $w(x_1, \dots, x_8)$. If G is MF, that image is trivial: $w \in \text{Res}_{\text{MF}}(E)$ by Theorem A, and the residual is functorial (Lemma S10.2). The canonical tuple of E satisfies R by definition, and its distinguished word is nontrivial by Theorem A. $\square$

Corollary S13.3 (a basic clopen set of non-MF marked groups). *In the space of eight-generator marked groups — quotients of the free group F_8 , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of R and $w \neq 1$ form a nonempty clopen set consisting entirely of non-MF groups.*

Proof. Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group E by Theorem A, and Proposition S13.2 excludes every MF point. $\square$

S13.3. Undecidability of MF recognition. Lemma S10.9(1) and Theorem A make MF a Markov property of finitely presented groups: the trivial group is MF, and by subgroup heredity the finitely presented group E embeds in no MF group. The Adian–Rabin theorem [Ra] therefore applies, and classically MF recognition is undecidable.

Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_e for the group presented by the code e .

Corollary S13.4 (undecidability, given a halting reduction). *Suppose given a computable map e sending each instance i of the halting problem to a presentation code $e(i)$ such that $G_{e(i)}$ is MF exactly when i halts. Then no algorithm decides, from a finite presentation, whether the presented group is MF, and the set of presentation codes of non-MF groups is not recursively enumerable.*

Effectiveness of the coding is necessary: a map attaching a finitely presented group to each code need not be computable, and an uncomputable choice would make the statement true for reasons having nothing to do with groups.

The corollary reduces from the halting problem, whose undecidability and non-recursively-enumerable complement are classical, so the fact that some finitely presented group has unsolvable word problem is not needed: the reduction asks only for a computable map and a correctness equivalence, and does not require its source to be a word problem. Constructing such a map is the Adian–Rabin argument [Ra] run over a group built from a machine; classically one takes the word problem of a finitely presented group with unsolvable word problem and applies Adian–Rabin to the Markov property “is MF”. That construction fixes one finite presentation P_0 whose group has undecidable word problem, and gives a computable map taking a word w in the generators of P_0 — the instance varies, the presentation does not — to a finite presentation whose group is trivial — hence MF — when $w = 1$ in P_0 , and contains E — hence non-MF, by subgroup heredity — when $w \neq 1$. The words representing 1 form a recursively enumerable set which, the word problem of P_0 being undecidable, is not recursive; hence the words with $w \neq 1$ form a set that is *not* recursively enumerable. An enumeration of the

non-MF presentations would enumerate that set, making the word problem decidable. Finitely presented non-MF groups occur in infinite multiplicity, and finitely generated ones in continuum multiplicity. The groups $E \times \mathbb{Z}^k$ for $k \geq 0$ are finitely presented, pairwise nonisomorphic, and non-MF, each containing E (Lemma S10.9(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks; the number of homomorphisms to $\mathbb{Z}/2$ separates them as well. Beyond finite presentation, pairing E with a continuum of pairwise nonisomorphic finitely generated groups yields $2^{\aleph_0}$ pairwise nonisomorphic finitely generated non-MF groups: each product $E \times N$ is non-MF, and a fixed countable group X satisfies $X \cong E \times N$ for at most countably many of the continuum many isomorphism types N — each such N embeds in X as a finitely generated subgroup, a countable group has only countably many finitely generated subgroups, and the types N are pairwise nonisomorphic. A map with countable fibres from a continuum-sized family has continuum-sized image. Products of pairwise nonisomorphic groups with a fixed factor can still be isomorphic, so it is the isomorphism types of the products that are counted. The deduction uses only that the N are finitely generated; a continuum of pairwise nonisomorphic finitely generated groups is given by the lamplighters over coset spaces of the free group, or by Neumann’s two-generator groups [Ne].

S14. LIMITATIONS OF THE OPERATOR-NORM METHOD

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm. Both proofs of Theorem 3.4 use that operator-norm control: without it the classes $[\text{Ad } U_n(g)]_\omega$ need not be multiplicative, and the spectral estimates of Section S4 are unavailable. The failure is exact and dimension-free: against the operator-norm estimate $\|\text{Ad}(U) - \text{Ad}(V)\| \leq 2 \|U - V\|$, there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints stay 2 apart in operator norm — the same constant, in the opposite direction.

Conversely, amplification by an identity block separates the two normalizations exactly. Padding a model with an identity block preserves every operator-norm quantity — the multiplicative defects and the pairwise separations are unchanged — while the normalized Hilbert–Schmidt distances decay with the padding ratio: below any prescribed bound, every approximate unitary representation can be replaced by one all of whose pairwise Hilbert–Schmidt distances are smaller than the bound. 2-norm triviality of an element therefore does not by itself prove a group non-MF: an obstruction must produce an invariant corner with its own normalized trace, so that the normalization is unchanged by amplification. A normalized trace taken from the ambient blocks does not survive amplification. The relative rank of that substructure is unconstrained: padding leaves the averaging corner $q_n M_{d_n}(\mathbb{C}) q_n$ untouched while d_n grows, so r_n/d_n may tend to zero while the

argument uses the corner's own normalized trace; the spectral projection q_n of Theorem S6.6 is normalized on its own range in the same way, and the collapse argument of Theorem S11.2 normalizes by the rank k_n .

Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]; that question is independent of Theorem 3.4. The adjoint spectral estimates used here need operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply, so they do not settle soficity or hyperlinearity of E or of W ; both groups are sofic by Theorems D and E.

Finite dimensionality of the coordinates cannot be replaced by stable finiteness of the coordinate algebras together with faithful traces. For faithfully traced unital C^* -coordinates A , the left and right actions on $L^2(A, \tau)$ generate a von Neumann algebra that need not be finite: for an infinite-dimensional II_1 factor in standard form it is all of $B(L^2(A, \tau))$. Constant coordinates $A_n = C_r^*(E)$ give a counterexample to any such generalization: the left regular representation embeds E exactly and injectively into the norm quotient

$$\prod_n A_n / \bigoplus_n A_n,$$

the element c centralizes the Kazhdan base there, and conjugation by t produces tct^{-1} , whose commutator with the base is the element u with $u^2 = w \neq 1$ of Theorem A. Precisely: Theorem 3.4 becomes false when its matrix coordinates are replaced by stably finite unital C^* -algebras carrying faithful tracial states and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

Questions. Every finite subgroup of a torsion-free group is trivial, so Theorem S6.3 is empty for such a group. In Theorem S6.6 a normal property-(T) subgroup of N_{conj} obstructs homomorphisms to $\mathcal{U}(\mathcal{Q})$ with no torsion anywhere. What remains for a torsion-free example is group-theoretic — a torsion-free finitely presented group whose subgroup N_{conj} contains a normal subgroup with property (T) — and Remark 5.2 gives a small-cancellation route to such a group, over the torsion-free finitely presented Kazhdan group of [FFF], which removed the analogous torsion obstacle for nonsoficity.

- (1) What is $\text{Res}_{\text{MF}}(E)$? It properly contains $\{1, w\}$ by Theorem B. Write $q_1: E \rightarrow E/\langle w \rangle$, let $D = D_{\text{coll}}(\bar{L}, q_1(t))$ for the witness $\bar{d}$ of that proof, and let $q_2: E/\langle w \rangle \rightarrow (E/\langle w \rangle)/D$. By Corollary S11.7 applied to $E/\langle w \rangle$ and then Corollary S10.5,

$$\text{Res}_{\text{MF}}(E) = q_1^{-1} q_2^{-1} (\text{Res}_{\text{MF}}((E/\langle w \rangle)/D)),$$

so by Corollary S10.4 the residual equals $q_1^{-1}(D)$ exactly when $(E/\langle w \rangle)/D$ is MF. Whether it is, is open, and neither collapse theorem applies to it: w is already trivial in $E/\langle w \rangle$, and the commutators generating D are already trivial in the second quotient. By Theorem 3.3, $\mathfrak{D}(E, \iota(\mathcal{B}))$ lies in the kernel of every finite-dimensional

linear representation, and hence of every homomorphism to a finite group; but after both quotients that conclusion has no content unless some element of $\mathfrak{D}(E, \iota(\mathcal{B}))$ has a nontrivial image there, and no such element is exhibited here. Deciding the question this way therefore requires producing one.

- (2) Is there a torsion-free finitely presented non-MF group? The analytic obstruction no longer requires torsion (Theorem S6.6), so the question is which source to apply it to.

The source of Theorem 5.1 cannot serve, because its relation $c^2 = 1$ makes the conjugation defect vanish in every torsion-free quotient. If $q: E \twoheadrightarrow Q$ has Q torsion-free then $q(c)^2 = 1$ forces $q(c) = 1$, hence $q(d) = 1$ for $d = tct^{-1}$, hence q kills every generator $[d, \iota(\lambda)]$ of N_{conj} . A quotient of that source with $q(N_{\text{conj}}) = Q$ and Q torsion-free is therefore trivial.

Take instead the one-sided Kazhdan datum with c of infinite order: for Γ torsion-free finitely presented with property (T) [FFF], $\alpha: \Gamma \hookrightarrow \Gamma$ injective and proper, and $a \in \Gamma \setminus \alpha(\Gamma)$, put

$$E_\infty = \langle \Gamma, t, c \mid t\gamma t^{-1} = \alpha(\gamma), \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma) \rangle,$$

with no relation on c . Viewing E_∞ as an HNN extension of $\Gamma \times \langle c \rangle \cong \Gamma \times \mathbb{Z}$ that identifies Γ with $\alpha(\Gamma)$, a finite-order element fixes a vertex of the Bass–Serre tree and so is conjugate into $\Gamma \times \mathbb{Z}$, which is torsion-free; hence E_∞ is torsion-free.

Set $d = tct^{-1}$ and $u = [d, a]$, a generator of N_{conj} . Then u has infinite order. Map E_∞ to the $\mathbb{Z}$ -lamp group $(\bigoplus_{G/\Gamma} \mathbb{Z}) \rtimes G$ over $G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$ by sending c to the lamp at the coset Γ ; the relation $[c, \gamma] = 1$ holds because Γ fixes that coset. Then $d \mapsto e_{t\Gamma}$ and $ada^{-1} \mapsto e_{at\Gamma}$, and $a \notin \alpha(\Gamma)$ gives $at\Gamma \neq t\Gamma$, so

$$u \mapsto e_{t\Gamma} - e_{at\Gamma} \neq 0,$$

of infinite order in $\bigoplus_{G/\Gamma} \mathbb{Z}$. This is the torsion-free counterpart of the central sign: there the lamp group is $\text{Cl}(X)$ and $u^2 = -1$; here it is $\bigoplus_X \mathbb{Z}$ and u is a difference of two distinct basis vectors.

What is wanted is a quotient $q: E_\infty \twoheadrightarrow Q$ with Q finitely presented, Q Kazhdan, Q torsion-free, Q two-generated, $q(N_{\text{conj}}) = Q$, and $q(u) \neq 1$. The first five hold for the trivial quotient, whatever the underlying data, so they carry no information on their own.

Given such a q with $Q \neq 1$, the group Q is two-generated, finitely presented, torsion-free, Kazhdan, and equal to its own MF residual, so every homomorphism from Q to the unitary group of a matrix quotient is trivial, and every nontrivial quotient of Q likewise fails to be MF; in particular Q itself is not MF. By Theorem S11.8, a group that equals its own collapse defect admits no nontrivial homomorphism to any countable MF group.

Hull’s small-cancellation theorem [Hu, Theorem 1.5] would supply such a q from a two-generated *suitable* subgroup S of N_{conj} — one whose induced action is non-elementary and which normalizes no finite subgroup. The quotient it returns is obtained by adjoining finitely many relators and restricts to a surjection from S ; property (T) descends from a torsion-free Kazhdan lattice through a common quotient. Torsion-freeness of Q follows from its torsion clause, which produces every finite-order element of the quotient from a source element of the same order, and E_∞ is torsion-free. The condition $q(u) \neq 1$ follows from injectivity on a prescribed ball, taking the radius large enough for that ball to contain u .

What is not available is a suitable such S . Lemma 5.8 of [Hu] produces one from an infinite-order h with $\langle h \rangle$ hyperbolically embedded, and infinite order alone does not give that, so u does not yet suffice. The Bass–Serre action of E_∞ does not supply it either: with vertex group $\Gamma \times \mathbb{Z}$ both associated subgroups are proper, and segments along the t -axis have stabilizers containing $\alpha^n(\Gamma)$, so that action is not acylindrical.

- (3) Does MF imply hyperlinearity? A faithful embedding into a matrix quotient need not make any coordinate trace faithful, so the defining MF representation does not immediately give a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.
- (4) By Proposition S10.7, every family of theorems placing subgroups in the MF residual generates, by iteration through quotients, a closure contained in Res_{MF} . For which groups do the criteria of Theorems S6.3, S6.6, S11.2, and S11.4 compute the residual exactly?
- (5) Does the obstruction survive when the matrix blocks are replaced by other building blocks? The proof of Theorem 3.4 uses finite dimensionality of $M_{d_n}(\mathbb{C})$ twice: for the conjugation action on $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, and for finiteness of the norm ultraproduct. Section S14 shows that faithful traces alone do not replace it. For which separable C^* -algebras A_n of real rank zero does every homomorphism $H \rightarrow \mathcal{U}(\prod_n A_n / \bigoplus_n A_n)$ still kill w ? The lifting of Lemma 2.1, which is where the matrix structure first enters, is a semiprojectivity question in the sense of Loring [Lo98].

S15. THE MAXIMAL GROUP C^* -ALGEBRA

For a countable group H , the maximal group C^* -algebra $C_{\text{max}}^*(H)$ is the completion of the group ring $\mathbb{C}[H]$ in the norm

$$\|x\|_{\text{max}} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of H ; the supremum is finite because $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$. Write u_h for the canonical unitary of $h \in H$.

Proposition S15.1 (the universal property). *The canonical map $h \mapsto u_h$ is injective, and for every unital C^* -algebra B lying in a fixed universe that contains H , and every homomorphism $\rho: H \rightarrow \mathcal{U}(B)$, there is a unique unital $*$ -homomorphism $C_{\max}^*(H) \rightarrow B$ with $u_h \mapsto \rho(h)$ for every $h \in H$.*

Injectivity comes from the left regular representation, uniqueness from density of $\mathbb{C}[H]$. Restricting B keeps “all representations” a set rather than a proper class, and every target below is a unitary group of a separable C^* -algebra.

Under a proper one-sided conjugation the maximal algebra contains an isometry that is not unitary, while the reduced algebra of E is stably finite.

Proposition S15.2 (a strictly dominated projection gives a proper isometry). *Let A be a unital C^* -algebra containing a projection p and a unitary u such that the conjugate upu^* satisfies $p(upu^*) = (upu^*)p = p$, equivalently $p \leq upu^*$, and $upu^* \neq p$. Then A contains an isometry that is not a unitary; consequently A is not stably finite and has no faithful tracial state.*

Remark S15.3 (the maximal algebra under proper one-sided conjugation). Let G contain a property-(T) subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$; the base of E is such a pair. Let $P \in C_{\max}^*(G)$ be the spectral projection at 1 of the Hermitian Kazhdan average $\frac{1}{2|S|} \sum_{s \in S} (u_s + u_s^*)$, which exists by the spectral gap of Section 3. Then $P \leq u_t P u_t^*$, and the inequality is strict: in the quasi-regular representation on $\ell^2(G/t\Gamma t^{-1})$ the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of Γ outside it, so the two projections differ there. By Proposition S15.2, $C_{\max}^*(G)$ is then not stably finite, has no faithful trace, and in particular is neither residually finite-dimensional nor MF. For the reduced algebra of E , by contrast, the canonical trace is faithful and the algebra is stably finite (Theorem C), so its failure of MF is not a failure of finiteness.

S16. THE THIRTEEN-RELATOR PRESENTATION OF $\mathrm{SL}_3(\mathbb{Z})$

Property (T) of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ is classical [BHV, Example 1.7.4(i)], and the presentations go back to [CRW, Theorem 2]. Proposition S1.1 derives property (T) for the twenty relators of (8) through the thirteen-relator presentation P_{13} , using the isomorphism $P_{13} \cong \mathrm{SL}_3(\mathbb{Z})$ proved here.

Theorem S16.1 (completeness of the thirteen-relator presentation). *The canonical homomorphism $P_{13} \rightarrow \mathrm{SL}_3(\mathbb{Z})$, sending each generator e_{ij} to the corresponding elementary matrix, is an isomorphism.*

Proof. Write $\pi: P_{13} \rightarrow \mathrm{SL}_3(\mathbb{Z})$ for the map in question, and

$$\omega_{12} = e_{12}e_{21}^{-1}e_{12}, \quad \omega_{13} = e_{13}e_{31}^{-1}e_{13}, \quad \omega_{23} = e_{23}e_{32}^{-1}e_{23}$$

for the three signed swaps; ω_{12} is the Weyl element of the upper-left 2×2 block, and the thirteenth relator is $\omega_{12}^4 = 1$. Let e_3 be the third basis column of $\mathbb{Z}^3$, let $\|\cdot\|_1$ be the ℓ^1 norm on $\mathbb{Z}^3$, and put

$$H_{\text{par}} = \langle e_{12}, e_{21}, e_{31}, e_{32} \rangle \leq P_{13},$$

the subgroup on the four root letters whose matrices fix e_3 .

Surjectivity. Division with remainder in the first column of $A \in \text{SL}_3(\mathbb{Z})$ is a product of integer powers of elementary matrices, and iterating it on columns and then on rows reduces A to a diagonal sign matrix. Each such sign matrix is the square of a signed swap, and each signed swap is the displayed three-transvection word.

Injectivity: a normal form. Write an element of P_{13} as a word $W = \ell_k \cdots \ell_1$ in the letters $e_i^{\pm 1}$. Given a column $b \in \mathbb{Z}^3$, the *trajectory* of W over b is the sequence $b, \pi(\ell_1)b, \pi(\ell_2\ell_1)b, \dots, \pi(W)b$, and W is *norm-monotone* over b when the ℓ^1 norms along that sequence do not decrease. Using only the thirteen relators, every W can be rewritten as

$$W = Gm, \quad m \in \langle \omega_{13}, \omega_{23} \rangle, \quad G \text{ norm-monotone over } b := \pi(m)e_3.$$

The rewriting is a descent, in the lexicographic order on $\mathbb{N} \times \mathbb{N}$, of the pair consisting of the largest ℓ^1 norm attained along the trajectory at a position where monotonicity fails, and the number of positions at which it attains that norm and fails. One step splits W at such a position and rewrites the offending pair of adjacent letters; the thirty-six ordered pairs of root families fall into four classes — cancelling, commuting, Steinberg, and braid — each handled by one of the displayed relators, and each strictly decreasing the pair. The thirty-six cases are enumerated in the supplementary material.

Injectivity: a kernel word has constant trajectory. Suppose $\pi(W) = 1$. The subgroup $\langle \omega_{13}, \omega_{23} \rangle$ maps into the signed permutation matrices, so $b = \pi(m)e_3$ is a signed basis column and $\|b\|_1 = 1$. The trajectory of G over b starts at b and ends at $\pi(G)b = \pi(m)^{-1}b = e_3$, again of norm one, and it is nondecreasing; hence it is constant. A root letter that preserves the ℓ^1 norm of a signed basis column fixes that column, so every letter of G fixes b — and, taking the whole word, $\pi(m)e_3 = \pi(G)b = e_3$, so $\pi(m)$ fixes e_3 .

Injectivity: reduction to the parabolic subgroup. Conjugation by ω_{13} and by ω_{23} permutes the twelve letters $e_i^{\pm 1}$, by two explicit tables derived from the relators. Applying them letter by letter turns $m^{-1}Gm$ into a word in root letters, each of which fixes $\pi(m)^{-1}b = e_3$. The root letters whose matrices fix e_3 are exactly $e_{12}^{\pm 1}, e_{21}^{\pm 1}, e_{31}^{\pm 1}, e_{32}^{\pm 1}$, so $m^{-1}Gm \in H_{\text{par}}$.

Injectivity: the subgroup $\langle \omega_{13}, \omega_{23} \rangle$. The group $\langle \omega_{13}, \omega_{23} \rangle$ acts on the six signed basis columns, and the orbit of e_3 is all six of them, with coset representatives

$$1, \quad \omega_{13}, \quad \omega_{23}, \quad \omega_{13}^2, \quad \omega_{13}^3, \quad \omega_{23}\omega_{13}^2$$

carrying e_3 to $e_3, e_1, e_2, -e_3, -e_1, -e_2$ respectively. Twelve Schreier relations, one for each generator and coset, have the form $\omega_{\bullet} r_j = r_{j'} \omega_{12}^k$: every Schreier

element is a power of the block Weyl element ω_{12} ; the twelve are listed in the supplementary material. The remaining Schreier relation is $\omega_{23}^2 = \omega_{12}^2 \omega_{13}^2$. In $K_2(3, \mathbb{Z})$ the Steinberg symbol $\{-1, -1\}$ is ω_{12}^4 , so this relation holds in P_{13} exactly because the thirteenth relator sets $\omega_{12}^4 = 1$. Hence every element of $\langle \omega_{13}, \omega_{23} \rangle$ is $r_j \omega_{12}^k$; if in addition its matrix fixes e_3 , then $r_j e_3 = e_3$ forces $j = 0$, leaving $m = \omega_{12}^k \in H_{\text{par}}$. With the previous step,

$$W = m \cdot (m^{-1} G m) \in H_{\text{par}}.$$

The next step is applied to this product and not to $m^{-1} G m$ alone, whose image under π is $\pi(m)^{-1}$ rather than 1.

Injectivity: the parabolic subgroup. Let $N = \langle e_{31}, e_{32} \rangle$ and $\Lambda = \langle e_{12}, e_{21} \rangle$. The Steinberg relations make N abelian and normalized by Λ , and every element of H_{par} is $e_{31}^a e_{32}^b \gamma$ with $\gamma \in \Lambda$. On Λ the displayed relators reduce to

$$\Lambda = \langle p, q \mid pqp = qpq, (pqp)^4 = 1 \rangle, \quad p = e_{12}, \quad q = e_{21}^{-1},$$

and π carries p and q to the two elementary matrices of the upper-left $\text{SL}_2(\mathbb{Z})$ block, whose entries outside that block are those of the identity. Put

$$\chi = pqp, \quad \psi = pq, \quad \eta = \chi^2.$$

Since $p = \psi^{-1} \chi$ and $q = \chi^{-1} \psi^2$, the pair χ, ψ generates Λ , and the braid relation is equivalent to $\chi^2 = \psi^3$; thus $\Lambda = \langle \chi, \psi \mid \chi^2 = \psi^3, \chi^4 = 1 \rangle$, and $\eta = \chi^2 = \psi^3$ is central with $\eta^2 = 1$. Killing η leaves $\langle \chi, \psi \mid \chi^2 = \psi^3 = 1 \rangle = \mathbb{Z}/2 * \mathbb{Z}/3$. The Möbius ping-pong argument identifies $\text{PSL}_2(\mathbb{Z})$ with the free product of the order-two and order-three subgroups generated by the images of $\pi(\chi)$ and $\pi(\psi)$, so the induced surjection $\Lambda / \langle \eta \rangle \rightarrow \text{PSL}_2(\mathbb{Z})$ carries each free factor isomorphically onto the corresponding one and is therefore an isomorphism. Hence $\ker(\pi|_{\Lambda}) \subseteq \langle \eta \rangle$; and $\pi(\eta) = \text{diag}(-1, -1, 1) \neq I$, so $\ker(\pi|_{\Lambda}) = 1$. Now let $W = e_{31}^a e_{32}^b \gamma \in H_{\text{par}}$ have $\pi(W) = 1$. The upper-left 2×2 block of $\pi(W)$ is the matrix of γ , so $\gamma = 1$; the bottom row of $\pi(W)$ is then $(a, b, 1)$, so $a = b = 0$. Hence $W = 1$, and $\ker \pi$ is trivial. $\square$

Remark S16.2 (the classical identification). The eight relations (7) present $\text{SL}_3(\mathbb{Z})$: Theorem S16.1 composed with the surjection of Step 2 identifies $\mathcal{R}$ with $\text{SL}_3(\mathbb{Z})$, and the base $\mathcal{B}$ is then the affine group $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ — the surjection $\mathcal{B} \rightarrow \bar{\Gamma}$ of Lemma S2.2 onto the matrix realization is an isomorphism. The presentations go back to [CRW, Theorem 2], with the matrix-labelled generators also recorded in [CLV], and property (T) of the affine group is classical [BHV, Example 1.7.4(i)].

ACKNOWLEDGEMENTS

We thank Caleb Eckhardt for discussions clarifying the finite-quotient analogy that organizes Section 1. We thank Francesco Fournier-Facio for formulating the group-theoretic input as a Kazhdan group that is not co-Hopfian, together with its ascending HNN extension. We also thank him for the small-cancellation construction recorded in Remark 5.2. We further thank him for helpful discussions on the distinct roles of the operator and

Hilbert–Schmidt norms. We thank Alon Dogon for pointing out that the interaction between those two norms also occurs in the earlier approach of Bachner–Dogon–Lubotzky [BDL].

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